Reading Financial Conditions in Units of Monetary Policy

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Abstract

How many points of monetary policy is a movement in financial conditions worth? An event study answers by asking how large a conventional policy surprise the movement resembles. A spending calibration asks how large a change in the policy rate would have a comparable effect on GDP. I compare four calibrations of the 10-year yield, equity valuation and the dollar on common series and one scale: market weights from two measures of policy news in FOMC announcement windows, the spending weights of the FRB/US model, and the published Goldman Sachs weights. Knowing how an index was built tells a reader surprisingly little about which other index it will resemble. The futures-based market weights and the published Goldman Sachs weights read a dollar appreciation relative to an equity decline almost identically, while the market weights based on the two-year yield and FRB/US give the dollar more than twice the relative weight. Over 1996–2026 the four readings of annual changes usually give the same directional signal, and materially opposing readings are rare. The size of a tightening is more sensitive. Switching between two implementations of the same method can change the reading as much as switching from market responses to spending effects, and the two closest readings come from different methods. GDP growth provides little evidence for choosing between the futures-based market reading and the FRB/US reading, and the unemployment comparison is sensitive to the sample.

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1 Introduction

Suppose the Federal Reserve leaves the policy rate unchanged. In one month stock prices fall; in another the dollar appreciates. Both movements tend to restrain spending. How much tightening has each delivered? One way to answer is to ask how large a monetary policy surprise would ordinarily produce a comparable movement in financial markets. Another is to ask how large a change in the policy rate would have a comparable effect on spending. A policymaker might reasonably expect the two answers to be close.

Policymakers face this question routinely. In November 2023 the Chair noted that "Financial conditions have tightened significantly in recent months, driven by higher longer-term bond yields, among other factors" (Powell, 2023), and in March 2024 that "there are many different financial conditions indicators, and you can kind of, you know, see different answers to that question" (Powell, 2024). A statement that financial conditions have tightened by the equivalent of a quarter point is a statement about policy only to the extent that a point of the index means a point of policy.

Figure 1 states the puzzle. It applies four calibrations to the same two movements: a 5 log point fall in equity valuation and a 3.9 log point appreciation of the dollar. Two of the calibrations are market-based. They weight asset prices by how they respond to monetary policy news in announcement windows, with the news measured by a factor from interest-rate futures or by the change in the two-year Treasury yield. Two are spending-based. They weight asset prices by their effects on GDP, in the Federal Reserve's FRB/US model and in the published weights of the Goldman Sachs index. I chose the sizes of the movements so that the first market calibration rates them alike. The published Goldman Sachs weights do too. The other two calibrations, one market-based and one spending-based, read the dollar appreciation as more than twice as tight as the equity decline. The difference between asking about policy news and asking about spending is real, and it does not predict the answer.

This paper asks three questions about that figure. What does a point of financial conditions mean under each method, and under what conditions would the two methods give the same weights? Why do these four calibrations weight equities and the dollar differently? And when does the choice change the reading of history?

The first answer is that agreement would be a coincidence. An asset's weight in a market-based index depends on how reliably it reveals policy news: how strongly it responds, and how much it moves for other reasons. Its weight in a spending-based index depends on what its movements do to consumption, investment and trade. Even if policy worked entirely through these asset prices, nothing would require the assets that best reveal a policy surprise to be the assets that matter most for spending. The two sets of weights give the same reading when asset prices move in the proportions that policy news produces. They can differ when asset prices move in other proportions, and they can also agree then, if the differences in weights offset.

The second answer has two parts that the data can trace. On the same announcements, the part of the two-year yield change that the futures factor misses moves long yields and the dollar and leaves equities unchanged, so a market index built to recover the two-year yield gives equities

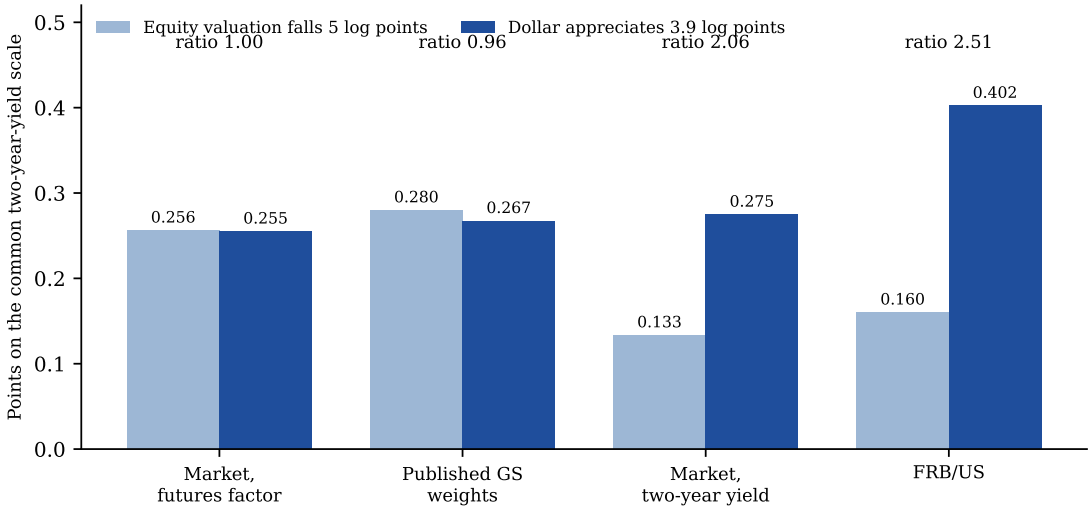


Figure 1: Two Tightenings Read by Four Calibrations

Notes: Readings of each movement under four portfolios of the same three assets relative to r^*, each normalized to read one on the response of the three assets to one unit of policy news measured by the futures factor (points on the common two-year-yield scale). Market: weights from announcement-window responses, with policy news measured by the futures factor or by the change in the two-year yield. FRB/US: effects on GDP after four quarters of persistent shifts. Published GS weights: the U.S. weights on the 10-year yield, equities and the dollar reported by Caballero and Simsek (2026), applied to the series used here with the GS policy-rate and spread terms held fixed. The ratio above each pair is the dollar reading over the equity reading; it does not depend on the normalization. The sizes of the movements were chosen to give equal readings under the futures factor; the ratio of one calibration's ratio to another's does not depend on that choice.

about half the weight. In FRB/US, a dollar appreciation reaches GDP almost entirely through net exports and an equity decline through consumption, and the model gives the dollar a larger weight relative to equities than the market responses do. The published Goldman Sachs weights, which are also calibrated to spending effects, are close to the futures-factor market weights on all three assets. That resemblance is a finding of the paper. Its cause is open.

The third answer comes from applying the four calibrations to the annual changes in the same three assets from 1996 to 2026. Three episodes show what the weights do. In 2008–09 every calibration registered a large tightening, and the choice of weights changed its size by more than half a point. In 1997–98 the dollar and equity valuations rose together, and one calibration read a tightening where the others read an easing. In 2022–24 falling valuations and a rising dollar both tightened conditions, the weights on them differed, and the readings nearly coincided because the differences offset. Over the full history the four readings usually give the same directional signal, and readings that oppose each other by a meaningful amount are rare. The size of a tightening is more sensitive. Switching between two implementations of the same method changes the reading about as much as switching from market responses to spending effects, and the two closest readings come from different methods.

Subsequent activity does not settle the choice. GDP growth provides little evidence for choosing between the futures-based market reading and the FRB/US reading, and the comparison for unemployment is sensitive to the sample.

Section 2 sets out what a point of financial conditions means under each method and when the methods would agree. Section 3 explains why the four calibrations weight the assets differently. Section 4 reads history with them. Section 5 asks whether subsequent activity can distinguish the readings, and Section 6 states the scope of the results and concludes.

1.1 Connection to the Literature

Weights and their meaning. Indexes of financial conditions differ mainly in the source of their weights (Hatzius et al., 2010): effects on output from a model (Dudley and Hatzius, 2000; Hatzius and Stehn, 2018; Ajello et al., 2023), principal components (Brave and Butters, 2011), or event-study regressions (Kuttner, 2001; Gürkaynak, Sack and Swanson, 2005; Bernanke and Kuttner, 2005; Nakamura and Steinsson, 2018; Bauer and Swanson, 2023). The older literature on monetary conditions indexes recognized that the weights on asset prices depend on the shock and the dynamics, so that their policy meaning is not invariant (Eika, Ericsson and Nymoen, 1996). I quantify that difficulty with four sets of U.S. weights on common assets and ask where it matters.

The closest comparisons. Arnaut and Bauer (2024) estimate how an existing index responds to policy surprises, Sack and Swanson (2023) relate FCI-G to policy-induced changes in the two-year yield, and Ajello et al. (2023) translate movements into a growth impulse with model multipliers. Bletzinger, Martorana and Mistak (2026) build a U.S. macro-finance index in units comparable to a policy rate and compare it with the Goldman Sachs index and FCI-G, finding divergences driven by equities and the strongest tightening in 2022–24. Comparisons of published indexes mix

the weights with differences in assets, transformations and units. My design holds the assets, the transformations and the policy unit fixed across calibrations, so that a difference in readings can be attributed to the weights. Boehm and Kroner (2025) show that announcements also move equities and exchange rates along a dimension orthogonal to yield changes, so a movement that departs from the conventional pattern can come from the announcement itself.

Information about activity. Hatzius et al. (2010) build an index conditional on past activity and inflation, Aramonte, Rosen and Schindler (2013) show that comparisons among indexes depend on the crisis in the sample, and Lombardi, Manea and Schrimpf (2025) separate safe yields from priced risk and find that risk predicts activity more strongly. Adrian, Boyarchenko and Giannone (2019) show that conditions shift the lower quantiles of growth, and Brave and Butters (2011) build an index of risk, credit and leverage. Caballero, Caravello and Simsek (2026) define FCI*, the level of financial conditions consistent with output at potential. The neutral rate matters for the level of every index (Laubach and Williams, 2003; Holston, Laubach and Williams, 2017, 2023; Hamilton et al., 2016; Bauer and Rudebusch, 2020; Kiley, 2020); I measure the inputs relative to r^* and do not estimate a neutral level for the index (Appendix B). **Why the meaning of a point matters.** Caballero, Caravello and Simsek (2024) study targeting an index of financial conditions in a closed economy, and Caballero and Simsek (2026) extend the analysis to a multi-asset open economy, in which domestic asset prices and the exchange rate both enter aggregate demand and policy responses link the markets. That gives a reason to care which mix of asset prices delivers a given index reading. My question comes one step earlier: when does a portfolio inferred from conventional policy news give those markets the same relative importance as a spending calibration? The answer does not require endorsing targeting. The same ambiguity arises for anyone who communicates about a desired path of financial conditions, since the path has to say what a point on it means, and for a neutral benchmark such as FCI* (Caballero, Caravello and Simsek, 2026), which takes the aggregation of asset movements as given.

2 What Does a Point of Financial Conditions Mean?

2.1 Three Asset Prices and Two Directions

I work with three asset prices throughout: the 10-year Treasury yield, equity valuation and the dollar. The 10-year yield is measured less expected inflation over ten years and less the neutral real rate r^* of Laubach and Williams (2003). Equity valuation is the gap between the cyclically adjusted earnings yield and r^*, expressed in log points of the price-earnings ratio, so that a fall of one log point is a fall of about one percent in prices at given earnings. The dollar is the log of the real broad dollar; it is a real exchange rate and is not measured relative to r^*. Measuring yields and valuations relative to r^* keeps a decline in the neutral rate from registering as an easing. Appendix B gives the construction.

A policy equivalent is a reading of a movement in these three prices that says how much tightening of monetary policy the movement amounts to. There are two natural ways to compute

one, and they run in opposite directions.

The first starts from policy and asks what it does to markets. Around a Federal Open Market Committee announcement, a surprise of size s moves the three asset prices by

$$x = \psi s + u. \tag{1}$$

The vector ψ is the usual response of the three prices to one unit of policy news, and one unit is the surprise that raises the two-year yield by one percentage point. The vector u collects everything else that moves the prices in the window, with covariance Σ. A market-based index reads a movement x as the size of the surprise that would ordinarily produce it. Its weights are the ones that best recover s from x.

The second starts from markets and asks what they do to spending. Let g_h be the effects on GDP, h quarters ahead, of a unit movement in each asset price that follows a specified path:

$$\Delta y_h = g_h' x. \tag{2}$$

A spending-based index reads a movement x as the change in the policy rate that would have the same effect on GDP. Its weights are the entries of g_h divided by the effect of a one-point change in the policy rate.

Neither reading is a claim about what caused the movement. A market-based reading of half a point says that the three prices moved as they would after policy news of that size; it does not say that policy news occurred. A spending-based reading of half a point says that the movement, if it lasts as assumed, will restrain GDP as much as a half-point rise in the policy rate; it does not say why asset prices moved. Both are translations of a financial movement into the units of policy, and both are useful to a policymaker for that reason: the first relates what markets did to what the central bank's own announcements usually do, and the second relates it to what the central bank is trying to achieve.

One equation describes how policy moves markets. The other describes how markets move spending. The four calibrations of this paper are two estimates of each: market weights with policy news measured by the futures-based factor of Nakamura and Steinsson (2018) and by the change in the two-year Treasury yield, and spending weights from the FRB/US model (Brayton, Laubach and Reifschneider, 2014) and from the published Goldman Sachs index.

2.2 When Would the Two Methods Agree?

Take the case most favorable to agreement. Suppose that policy affects spending only through these three asset prices, at the horizon and for the paths that the spending calculation assumes. A unit of policy news then lowers GDP by $g_h'\psi$. Would the assets that best reveal a policy surprise also be the assets that matter most for spending?

The weights answer the question. Put both indexes on a common scale, on which each reads one for the usual response ψ. The market weights that recover s with the smallest error are the

generalized least squares weights, and the spending weights are the spending effects rescaled:

$$w_M = \frac{\Sigma^{-1}\psi}{\psi'\Sigma^{-1}\psi}, \qquad w_{B,h} = \frac{g_h}{g_h'\psi}. \tag{3}$$

The two portfolios give the same reading for every possible movement in asset prices if and only if g_h is proportional to $\Sigma^{-1}\psi$. That condition ties together two things that have no reason to be tied. The right-hand side is about signal quality: how strongly each asset responds to policy news, and how much each moves for other reasons in the announcement windows. The left-hand side is about transmission: what a movement in each asset does to consumption, investment and trade.

Two examples show how they come apart. Suppose equities respond strongly to policy news but also move a great deal on earnings news that arrives in the same windows. Their independent contribution to recovering the policy surprise is then modest, and their market weight is small, even if a persistent fall in valuations has a large effect on consumption. Conversely, the dollar can respond to policy news so consistently that it is the best single indicator of a surprise, in an economy where trade is a small share of spending and the dollar matters little for GDP. Complete transmission through the three assets imposes no requirement that signal quality line up with spending effects. A market-based index and a spending-based index can both be right about their own question and disagree about a given movement.

The same reasoning applies within each method. Two market indexes that define policy news differently have different ψ and Σ, and two spending calibrations have different g_h. Nothing in the algebra makes the difference between methods larger than the differences within them. That is an empirical matter, and the answer in these data is that it is not.

2.3 Agreement in the History We Observe

Agreement for every possible movement is a demanding standard. What a policymaker needs to know is whether the readings agree for the movements that occur. Let x_t be the change in the three asset prices over some period, and $M_t = w_M'x_t$ its market reading. Split the movement into the part that policy news of size M_t would produce and a remainder:

$$x_t = \psi M_t + r_t. \tag{4}$$

The remainder r_t is the part of the movement that the usual policy response cannot reproduce. It is observable given the weights and the response vector. Computing it does not require knowing what moved markets in that period. For any two portfolios j and k on the common scale, the difference between their readings $I_{j,t}$ and $I_{k,t}$ is

$$I_{j,t} - I_{k,t} = (w_j - w_k)'r_t, \tag{5}$$

because both read one on ψ. The difference between two indexes prices only the remainder.

Three cases follow. First, all readings agree when asset prices move in the proportions of ψ,

whatever caused the movement. A fall in equity prices, a rise in long yields and an appreciation of the dollar in the usual proportions read as the same tightening under every calibration, whether a policy surprise, a change in risk appetite or foreign developments produced them. Second, two portfolios with similar weights give similar readings even when asset prices move in unusual proportions. Third, two portfolios with quite different weights can agree on a particular movement, because the remainder can load on the differences in weights with opposite signs. If equities fall and the dollar rises, a portfolio that gives equities less weight and the dollar more can arrive at the same reading as one that does the reverse.

The usual response ψ makes the cases concrete. Policy news that raises the two-year yield by one point raises the 10-year yield by about half a point, lowers equity prices by about 8 percent and raises the dollar by about 4 percent. A year in which equities fall 8 percent, the dollar rises 4 percent and long yields rise half a point is one point of tightening under all four calibrations. A year in which equities fall 16 percent and nothing else moves is not a multiple of that pattern. Its market reading is the policy news that best accounts for it, and its remainder is large: equities fell by more than that news would explain, and yields and the dollar by less. How the calibrations price that remainder is where they differ.

The third case has a consequence for how to read evidence. Two indexes can track each other for years without embodying the same economics, and agreement in a period says little about whether they will agree in the next. What matters is which asset prices move, and in which combinations.

2.4 Units and the Common Scale

Each method has its own unit. A market-based index is in points of the two-year yield: a reading of one is a movement that resembles the response to policy news that raises the two-year yield by one percentage point. A spending-based index is in points of the funds rate: a reading of one is a movement with the GDP effect of a persistent one-point rise in the federal funds rate. The published Goldman Sachs index has its own index points.

Comparing weights requires one scale, and I use the first. Each set of weights is divided by its own reading of the futures-factor response ψ. The conversion is itself informative for FRB/US. Under persistent asset paths, FRB/US assigns 0.954 funds-rate points (90 percent interval 0.86 to 1.08) to the asset response associated with a surprise that raises the two-year yield by one point. With a half-life of four quarters for the asset movements it assigns 0.82. The conversion depends on how long the movements are assumed to last, and the relative weights on the assets depend on it much less (Appendix D.4). After the division, every reading is in points on the common two-year-yield scale. Agreement of the four portfolios along ψ then follows from the units and is not evidence of anything. The evidence is in how they read the other combinations of asset movements that occurred.

Table 1: Four Sets of Weights on One Scale

	Market, futures factor	Published GS weights	Market, two-year yield	FRB/US
10-year yield (per percentage point)	0.609 [0.147, 1.114]	0.515	0.926 [0.476, 1.328]	0.590
Equity valuation (per log point)	−0.051 [−0.068, −0.030]	−0.056	−0.027 [−0.046, −0.011]	−0.032
Dollar (per log point)	0.065 [0.021, 0.112]	0.069	0.070 [0.033, 0.110]	0.103

Notes: Weights on the three assets relative to r^*, each set normalized to read one on the response of the three assets to one unit of futures-factor policy news (the common two-year-yield scale). Market weights: generalized least squares on 253 announcement windows, with policy news measured by the futures factor or by the change in the two-year yield; brackets are 90 percent intervals over 5,000 bootstrap draws of the announcements (for the two-year-yield weights, of the weights on their own signal, which differ from the entries by a factor of 0.999). FRB/US: effects on GDP after four quarters of persistent shifts. Published GS weights: the U.S. weights on the 10-year yield, equities and the dollar, with the GS policy-rate and spread terms held fixed. The FRB/US and GS weights are not estimated here.

3 Why Do the Four Calibrations Weight the Assets Differently?

Table 1 gives the four sets of weights on the common scale. The futures-factor market weights and the published Goldman Sachs weights are close on all three assets. The market weights built on the two-year yield put more weight on the 10-year yield and about half as much on equities. FRB/US puts the most weight on the dollar and, with the two-year-yield weights, the least on equities. For the movements of Figure 1, these weights produce dollar-to-equity readings of 1.00, 0.96, 2.06 and 2.51. This section explains what the data can explain about those differences: one mechanism on the market side and one on the spending side.

3.1 The Market Side: What Counts as Policy News

The market weights come from the 253 announcements from January 1996 to September 2026 in the San Francisco Fed's U.S. Monetary Policy Event-Study Database (Acosta et al., 2025). In a narrow window around each announcement I measure the changes in the 10-year yield, the S&P 500 and the dollar, and a measure of the policy news. The weights recover the news from the three changes, so they depend on what the researcher counts as news.

Two measures are natural. The first is the factor of Nakamura and Steinsson (2018), the common component of the changes in five interest-rate futures that span the next year or so of expected policy. The second is the change in the two-year Treasury yield in the same windows, which is the unit of the index. The two measures share most of their variation: their correlation is 0.90. They are not the same. Write the two-year change as the factor plus a remainder. The remainder is about 20 percent of the variance of the two-year change.

Figure 2 shows what the remainder moves. A point of the factor raises the 10-year yield by 0.53 points, lowers the S&P 500 by 7.78 percent and raises the dollar by 4.22 percent. A point of the remainder is associated with a rise of 0.80 points in the 10-year yield and of 4.30 percent in the dollar. It is associated with no discernible movement in equities: the estimate is 0.55 percent, with an interval from −5.14 to 5.23. The part of the two-year yield change that futures do not capture moves the yield curve and the dollar, and equity prices do not respond to it.

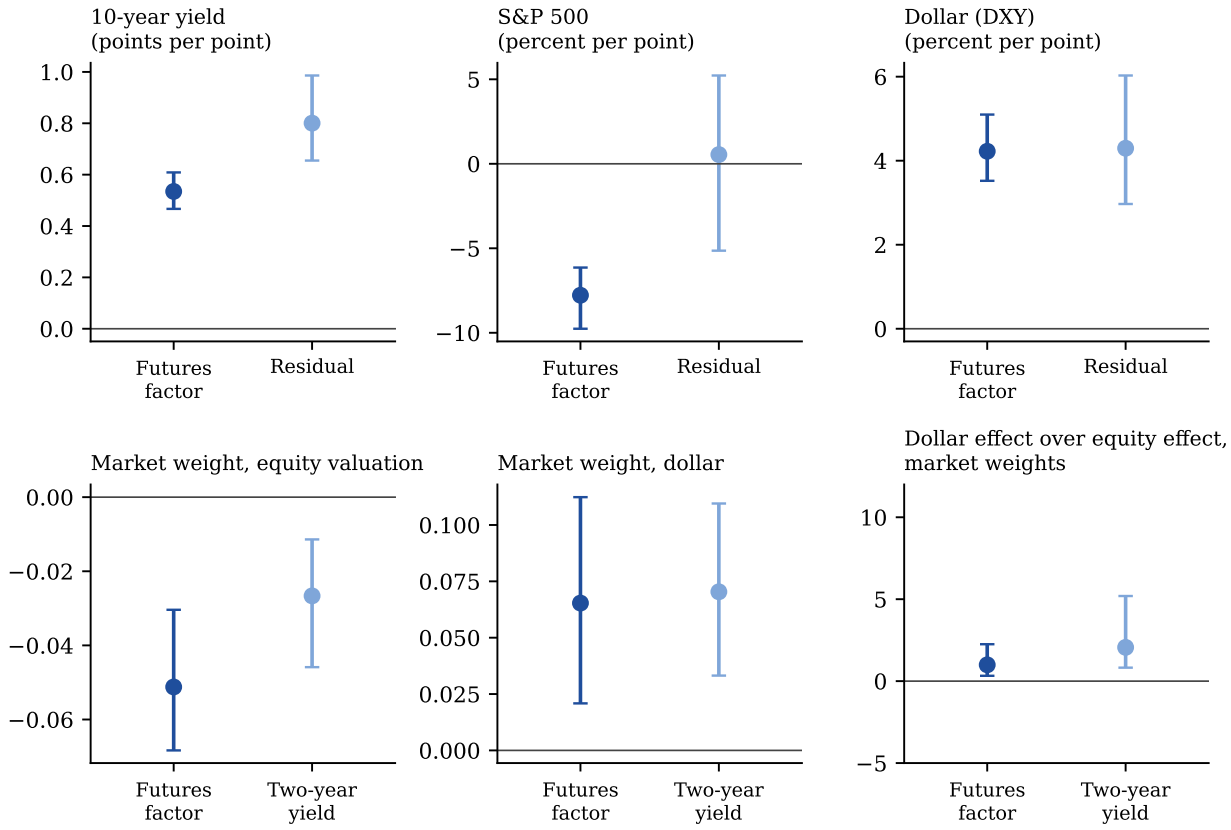


Figure 2: What the Two Measures of Policy News Capture

Notes: Top: responses of the 10-year yield (points), the S&P 500 and the dollar (DXY) (percent) in the announcement windows to one point of the futures factor and to one point of the part of the two-year yield change that is orthogonal to it; the factor is scaled to the two-year change. Bottom: the market weight on equity valuation and on the dollar, and the market's dollar-over-equity effect ratio, when policy news is measured by the futures factor and by the two-year yield. Bars are 90 percent intervals from 5,000 bootstrap draws of the announcements, drawn in common for the two signals.

That observation explains the weights through equation (3). When policy news is the futures factor, the remainder is part of u: it is noise that moves the 10-year yield and the dollar, and it makes those two assets less reliable indicators of the news. Equities, which do not respond to it, are the cleaner signal and receive a large weight. When policy news is the two-year yield, the remainder is part of the signal. The 10-year yield and the dollar now track the news more closely, and equities, which respond to only part of it, are the less informative asset. Both the response vector ψ and the covariance Σ change, and each accounts for about half of the change in the equity weight. The weight on equity valuation falls from -0.051 to -0.027, and the market's dollar-to-equity weight ratio roughly doubles, with a 90 percent interval for the change from 0.98 to 5.47 times.

The data do not say which measure is right. The remainder may be policy news about horizons that the futures do not cover, a movement in term premiums around announcements, or measurement error in a single yield. Each reading has a different implication for which index a policymaker should prefer, and the announcement responses cannot distinguish them. What they establish is that the answer to the question of Figure 1 depends on a choice that most users of a market-based index would not think of as consequential.

One caution applies to both market calibrations. The weights describe what the three asset prices do when policy moves. They do not describe what causes the asset prices to move at other times, and an announcement can also reveal the central bank's view of the economy, so that the responses mix policy and information (Nakamura and Steinsson, 2018). I measure the responses as markets deliver them. The index measures how much a movement resembles policy news, and it is not a forecast of activity.

3.2 The Spending Side: What Happens After Asset Prices Move

The spending weights answer a different question, and FRB/US lets a reader see the answer take shape. Figure 3 traces the two movements of Figure 1 through the model. Each asset price shifts persistently while the policy rate, the yield curve, credit spreads and the other asset price are held at their baseline paths.

The equity decline works through domestic demand. Lower equity wealth reduces consumption, which supplies 92 percent of the fall in GDP after four quarters, and a higher cost of capital reduces business investment. Weaker demand also reduces imports, so net exports offset part of the decline. The dollar appreciation works through trade. Net exports account for 94 percent of the fall in GDP, mostly through lower exports, and domestic demand hardly moves. After four quarters the equity decline has lowered GDP by 0.070 percent and the dollar appreciation by 0.177 percent. The model's answer to Figure 1 is therefore a statement about trade: an appreciation of 3.9 percent does more to GDP within a year than a fall of 5 percent in equity valuation, because in the model trade responds by more within a year than consumption responds to wealth.

Duration belongs in a spending calculation in a way that it does not in a market calculation. The effect of a movement on GDP depends on how long the movement lasts, and the weights in Table 1 assume that it persists. If both movements decay with a half-life of four quarters, FRB/US

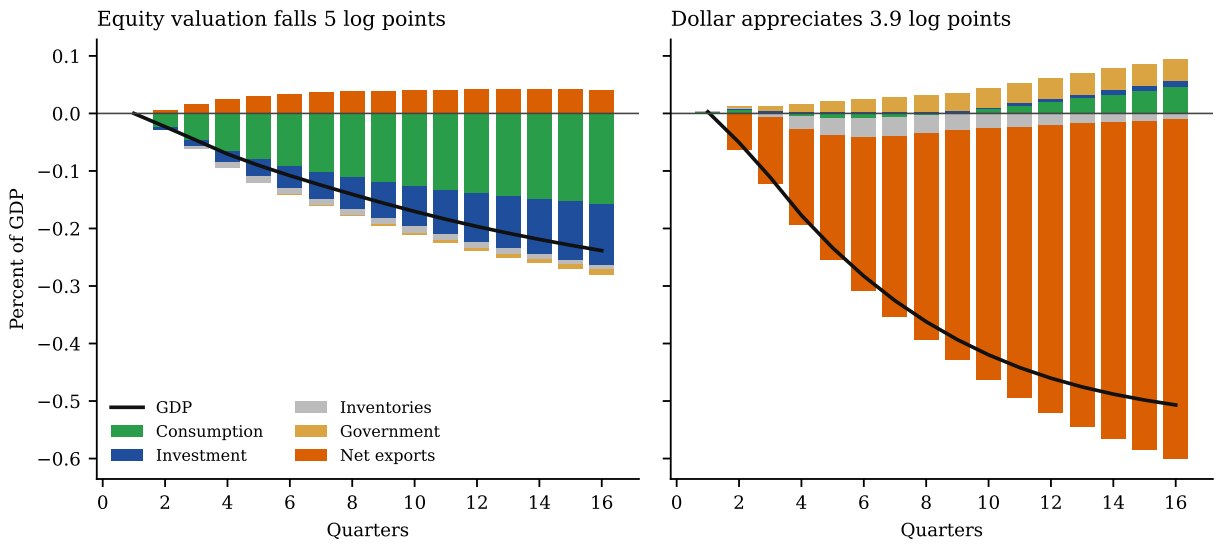


Figure 3: How the Two Movements Reach GDP in FRB/US

Notes: Percent deviation of real GDP from baseline and contributions by expenditure component, persistent shift in the variable, other financial variables held at baseline, policy rule inactive. Investment is residential and business fixed investment; net exports are exports less imports. The figure uses the Board's FRB/US package with the data vintage of October 2026, the vintage behind every FRB/US number in the paper.

assigns less tightening to each, and the conversion of Section 2.4 falls from 0.954 to 0.82. The ratio of the dollar reading to the equity reading barely changes, at 2.55 against 2.51. Persistence matters for how much tightening a movement represents and matters little for which asset counts more.

The experiment requires matching the market series to the model's variables, and the matches are approximate. The equity experiment raises the required return on equity, and household stock wealth falls with it, so that a fall of 5 log points in valuation is a fall of 5 percent in wealth. The market weights are estimated on the S&P 500 return and the DXY dollar index in the announcement windows and are applied to equity valuation and to the real broad dollar. A one percent return is treated as one log point of valuation, and the two dollar indexes as the same unit.

3.3 A Second Spending Calibration

If FRB/US were the only spending calibration, a reader might conclude that spending-based indexes weight the dollar more heavily than market-based ones. The published weights of the Goldman Sachs U.S. financial conditions index show that the conclusion does not follow. As Caballero and Simsek (2026) describe them, the weights are calibrated to each variable's estimated effect on real GDP growth one year ahead, and a one-point rise in the index is estimated to lower GDP by about 1 percent after four quarters. The U.S. weights are 4.4 on the policy rate, 45.1 on the 10-year yield, 39.6 on a corporate spread, 4.9 on an equity variable and 6.0 on a trade-weighted dollar. Caballero and Simsek (2026) illustrate them with an example close to Figure 1: a 20 percent fall in equity

prices and a 17 percent appreciation of the currency each tighten the index by about one point.

I apply the published weights on the 10-year yield, equities and the dollar to the same series as the other three portfolios and hold the policy-rate and spread terms fixed. The portfolio is the published GS weights on the common assets. It is not the Goldman Sachs index, whose series and transformations differ. On the common scale its weights are 0.515, −0.056 and 0.069, against 0.609, −0.051 and 0.065 for the futures-factor market index.

A spending calibration can therefore resemble a market portfolio closely. The two spending calibrations differ from each other by more: relative to equities, FRB/US gives the dollar about two and a half times the weight that the published GS weights do. The published documentation does not identify the feature of the underlying model that produces the difference, so the paper cannot explain it as it explains the other two. What can be said is that the difference lies where the expenditure channels of Figure 3 would put it, in how much a movement in the dollar does to GDP within a year relative to a movement in equity prices.

4 When Does the Choice Change the Reading of History?

4.1 Reading a Movement with Four Sets of Weights

I apply the four portfolios to the change over the preceding 12 months in the three asset prices, in each of the 369 months from January 1996 to September 2026. All four readings are on the common scale and are retrospective, with weights estimated on the full sample. They cover three asset prices. Corporate spreads, which the Goldman Sachs index and the full FRB/US calculation include, are held fixed, so the readings are not complete measures of financial conditions in a period such as 2008.

The difference between two readings is elementary accounting. For portfolios j and k,

$$I_{j,t} - I_{k,t} = \sum_{a} (w_{j,a} - w_{k,a})\, x_{a,t}, \tag{6}$$

where a runs over the 10-year yield, equities and the dollar. Each term is a difference in weights times the movement in one asset price. A difference in weights matters in proportion to how far the asset moved, and the terms can reinforce or offset each other. Figure 4 shows the four readings in three episodes, and each episode illustrates a different implication of equation (6).

4.2 2008–09: Agreement on Direction, Disagreement on Size

Equity valuations roughly halved between 2007 and March 2009. Measured over 12 months and relative to r^*, equity valuation fell by 34.39 log points a year on average from January 2008 to December 2009. The dollar changed little on average. The 10-year yield fell, and expected inflation and r^* fell by more, so the yield relative to r^* rose by 0.41 point a year.

One asset dominated the movement, so one difference in weights dominates equation (6). The futures-factor and GS portfolios give equities about −0.051 and −0.056 per log point, and the

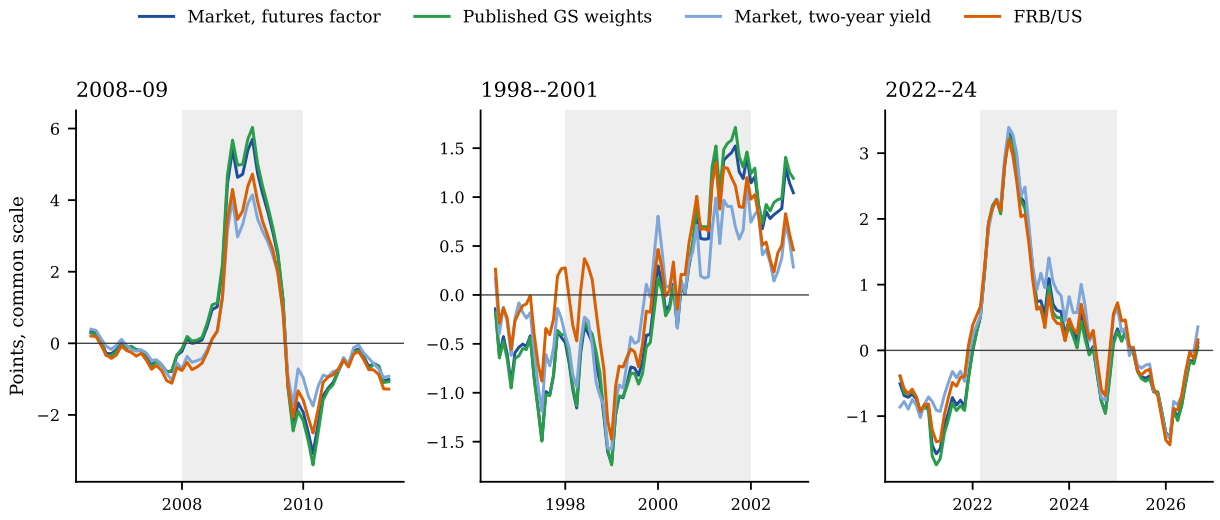


Figure 4: Four Readings, Month by Month

Notes: Twelve-month change in financial conditions under each portfolio, in points on the common two-year-yield scale, normalized to the futures-factor response. The shaded months are January 2008 to December 2009, January 1998 to December 2001 and March 2022 to December 2024. Corporate spreads are held fixed.

two-year-yield and FRB/US portfolios about −0.027 and −0.032. The fall in valuations alone contributes 1.76 points of tightening to the futures-factor reading and 1.10 to the FRB/US reading. The four readings average 2.05 and 2.17 points for the futures-factor and GS portfolios and 1.33 and 1.40 for the two-year-yield and FRB/US portfolios.

The month-by-month readings in Figure 4 show the same pattern at larger scale. The 12-month change peaked in March 2009, at 5.70 and 6.03 points under the futures-factor and GS weights and at 4.15 and 4.73 under the two-year-yield and FRB/US weights. At the widest, the highest and lowest of the four readings were 2.00 points apart. The futures-factor and FRB/US readings differed by at least a quarter point in 22 of the 24 months. The four lines in the figure nevertheless rise and fall together, and the correlations among the four readings over the full sample are between 0.93 and 1.00. High correlation is a statement about direction and timing. It is compatible with large differences in level.

Every calibration says that financial conditions tightened sharply. They differ by more than half a point of policy on how sharply. A policymaker asking in 2009 how much stimulus was needed to offset the tightening in these three markets would have received answers a half point or more apart, depending on a choice of weights that the direction of the movement did nothing to reveal. This is the case in which agreement on direction conceals a large difference in magnitude, and it arises whenever one asset price with a contested weight moves far.

4.3 1997–98: Offsetting Movements and Opposite Signs

The second case is the reverse. In the year to January 1998 the dollar appreciated by 12.79 percent in real terms, during the Asian financial crisis, equity valuations rose by 13.54 log points and the 10-year yield relative to r^* fell by 1.03 point. Two of the three movements eased financial conditions and one tightened them.

The sign of the reading then depends on the weights. Under the futures-factor market weights, the appreciation adds 0.84 points of tightening, the rise in valuations subtracts 0.69 and the fall in yields subtracts 0.63, for a net easing of 0.49. Under the FRB/US weights the same appreciation adds 1.32, the rise in valuations subtracts only 0.43 and the fall in yields subtracts 0.61, for a net tightening of 0.28. The published GS weights and the two-year-yield weights read an easing of 0.41 and 0.42.

An analyst watching household wealth and borrowing costs and an analyst watching the trade balance could have looked at the same year and given opposite accounts of it, and the four calibrations encode those two views in different proportions. The disagreement persisted through the first half of 1998 and faded as the dollar stopped rising. The same configuration returned in early 2015. The dollar had appreciated by about 11 percent over the year, valuations had risen by about 10 log points and long yields had fallen. In March and April 2015 FRB/US again read a tightening of about a quarter point while the other three read an easing of about a third of a point. Over the longer period from 1998 to 2001 the readings average -0.06, -0.03, -0.07 and 0.16, all small, because the boom in valuations and its reversal offset each other within the window. Opposite signs arise when asset prices move in opposing directions and the weights differ on the assets that moved most. Opposing movements of that size are unusual, which is why such months are rare.

4.4 2022–24: Different Weights, the Same Reading

The third case is the one that equation (5) predicts and that intuition resists. From March 2022 to December 2024 all three asset prices tightened conditions. The 10-year yield rose, equity valuations fell by 5.04 log points a year on average and the dollar appreciated by 2.85 percent a year. The four readings average 1.02, 0.97, 1.21 and 1.01 points. The futures-factor reading and the FRB/US reading differ by one hundredth of a point.

They agree for a reason that has nothing to do with similar weights. Relative to the futures-factor reading, FRB/US assigns 0.10 point less tightening to the fall in equities, because it gives equities less weight, and 0.11 point more to the appreciation, because it gives the dollar more. The two terms of equation (6) cancel. The same two differences in weights that separated the readings by more than half a point in 2008–09 and reversed their signs in early 1998 left them indistinguishable here, because equities and the dollar both moved in the tightening direction and by amounts that happened to balance.

The cancellation is a property of the averages. Month by month, the FRB/US reading less the futures-factor reading ranged from -0.32 to 0.40 point within the episode and was at least a quarter point in absolute value in 6 of the 34 months. At the peak of the tightening, in October 2022, the

Table 2: Four Readings of History

	Relative to r^* (paper)	Real, 100 ln CAPE	Nominal, 100 ln CAPE, nominal dollar
Mean absolute distance between two readings (basis points)			
Market, futures factor vs Published GS weights	7	8	10
Market, futures factor vs FRB/US	26	25	26
Market, two-year yield vs FRB/US	23	25	32
Market, futures factor vs Market, two-year yield	28	35	39
FRB/US vs Published GS weights	28	31	32
Market, two-year yield vs Published GS weights	34	43	49
Judgments across the four readings, 369 months			
All four the same sign (percent of months)	81.6	72.6	73.4
Months with materially opposing readings, 25 basis points	8	16	26
Months with materially opposing readings, 10 basis points	38	63	62
Four-reading range at least 25 basis points (percent of months)	70.5	75.1	78.3
Average reading, January 2008 to December 2009 (points)			
Market, futures factor	2.05	1.00	0.88
Published GS weights	2.17	1.15	1.06
Market, two-year yield	1.33	0.31	0.10
FRB/US	1.40	0.58	0.50

Notes: Twelve-month changes in the 10-year yield, equity valuation and the dollar, monthly, January 1996 to September 2026 (369 overlapping months), read with the four sets of weights of Table 1. Each column applies all four sets, unchanged, to the same series: the paper's series relative to r^*; the real 10-year yield, 100 times the log of the cyclically adjusted price-earnings ratio (prices over a 10-year average of earnings, the documented GS equity concept) and the real dollar; and the nominal 10-year yield, the same equity measure and the nominal broad dollar, the nearest publicly reproducible GS inputs. Distances are mean absolute differences between two readings in basis points of the two-year yield; the mean absolute futures-factor reading is 77 basis points in the first column. Materially opposing: at least one reading at least the threshold above zero and one at least the threshold below. Range: the largest less the smallest reading. Corporate spreads are held fixed. Intervals for the first column are in the text.

four readings of the 12-month change were between 3.21 and 3.39 points. Against a movement of that size the differences among calibrations were small.

The episode carries two lessons. Near-equality of two readings is not evidence that the indexes embody the same economics. And the robust statement about 2022–24 is that every calibration shows a tightening of about one point or more. Equality to the second decimal belongs to one pair of calibrations: the two-year-yield reading is 1.21, about two tenths higher, because that portfolio gives the 10-year yield more weight.

4.5 How Representative Are the Episodes?

Table 2 summarizes all 369 months. Its first column uses the series of the paper.

Distances. The closest pair crosses the two methods. The futures-factor market reading and the published GS weights differ by 7 basis points on average. The widest pairs include both within-method pairs: 28 basis points between the two market calibrations and 28 between FRB/US and the published GS weights, against 26 between the futures-factor market reading and FRB/US. The average absolute reading itself is 77 basis points, so a typical difference of 25 to 30 basis points is about a third of a typical movement. Switching between two implementations of the same method changes the reading about as much as switching from market responses to spending effects.

These are rankings of point estimates. The market weights are estimated, and over bootstrap draws of the announcements the difference between the futures-factor to FRB/US distance and the

distance between the two market calibrations is −3 basis points with a 90 percent interval from −19 to 27. The two are comparable in this sample, and the data cannot rank them. The futures-factor to GS distance has an interval from 6 to 44 basis points. A distance cannot be negative, so when two sets of weights nearly coincide, sampling error in one of them can only raise it, and the estimate sits at the bottom of its interval.

Direction. All four readings have the same sign in 81.6 percent of months (90 percent interval 70.2 to 82.9). Months in which two readings oppose each other materially are rare. With a threshold of 25 basis points, meaning at least one reading 25 basis points above zero and one 25 basis points below, there are 8 such months in 369, or 2.2 percent (1.6 to 8.1). Six of them are the months around January 1998 and two months in early 2015, when the dollar and valuations again rose together, and in all six FRB/US alone reads a tightening. The other two, in 2003 and 2008, are months in which valuations had fallen by more than 20 percent and the dollar had depreciated. With a threshold of 10 basis points the share rises to 10.3 percent. What counts as a meaningful disagreement is the policymaker's choice, and the conclusion that opposition is rare holds for thresholds a policymaker would care about.

Size. The largest and smallest of the four readings differ by at least 25 basis points in 70.5 percent of months. Disagreement about the size of a movement is the normal state of affairs, and disagreement about its direction is the exception. The first statement is less precise than the second. For the futures-factor and FRB/US pair alone, the share of months with a gap of at least 25 basis points is 36.6 percent, with an interval from 10 to 70 percent. The intervals reflect the estimation of the market weights and are conditional on the observed history and on the FRB/US and GS weights, which are not estimated here.

What the input series change. The comparison holds the series fixed across the four portfolios, which isolates the effect of the weights. A second question is how to measure the asset movement itself. The second and third columns of Table 2 apply the same four sets of weights to other series: the real 10-year yield and the log of the cyclically adjusted price-earnings ratio without the adjustment for r^*, and then the nominal yield and the nominal dollar, which are the nearest public counterparts of the Goldman Sachs inputs.

The ranking of the portfolios survives. The futures-factor and GS readings remain the closest pair, at 8 and 10 basis points, and each within-method distance remains at least as large as the futures-factor to FRB/US distance. The readings themselves change appreciably. Without the adjustment for r^* the tightening of 2008–09 is about half as large under every calibration, because the decline in the neutral rate during the crisis no longer counts as a tightening of yields and valuations relative to neutral. All four readings share a sign in 73.4 percent of months, down from 81.6, and months of material opposition are more frequent. How an asset movement is measured relative to neutral matters for the level of every reading and for how often the readings split. It matters much less for which portfolios resemble each other.

5 Can Subsequent Activity Distinguish the Readings?

5.1 The Question

If the readings differ, a natural response is to ask the data which is right. Consider two years in which the futures-factor market reading tightened by the same amount. In one, the dollar accounted for more of the tightening and equities for less, so that the FRB/US reading was higher. Did activity subsequently weaken more in that year? If the market reading is a sufficient summary of the three asset prices for predicting activity, the answer is no. If the FRB/US reading is sufficient, the answer is yes, and by a known amount.

The regression is

$$y_{t+h} - y_t = \cdots + \beta M_t + \gamma D_t, \tag{7}$$

where M_t is the 12-month change in the market reading, D_t is the FRB/US reading of the same asset changes less M_t, and the omitted terms are lags of the outcome. If the market reading is sufficient, $\gamma = 0$. If the FRB/US reading $M_t + D_t$ is sufficient, $\gamma = \beta$. The outcome closest to the spending calibration is real GDP growth over the next four quarters, since the FRB/US weights are effects on GDP at that horizon. I also report the change in the unemployment gap over the next 12 months. The test compares two of the four calibrations. The other two readings are close to these two in the dimensions the test can detect, and a test among four would have less power still.

5.2 The Answer

Table 3 reports the two restrictions. For GDP growth over 1976–2026 neither is rejected. The coefficient on composition is 0.87, with a 90 percent interval from −0.78 to 2.51, and its difference from the coefficient on the market reading is 1.03 (−0.39 to 2.44). The interval is wide in economic terms. Moving composition from its 5th to its 95th percentile at a given market reading changes predicted growth over the next year by 0.95 percentage point, with an interval from −0.85 to 2.76. The data are consistent with composition mattering a great deal in either direction and with its not mattering at all. The results from 1996 and without 2008–15 are similar.

The unemployment comparison depends on the sample. Over 1976–2026 neither restriction is rejected. From 1996 the FRB/US restriction is rejected at the 10 percent level: the coefficient on composition is −1.19, and its difference from the coefficient on the market reading is −1.19 (−2.31 to −0.07). Without 2008–15 the estimates are close to the FRB/US restriction instead. In economic terms, moving composition from its 5th to its 95th percentile over 1976–2026 changes the predicted change in the unemployment gap by −0.29 point, with an interval from −1.06 to 0.48. The sign of the composition coefficient changes across the three samples. I do not read the rejection as evidence for the market weights. It treats the market weights as known, it is one of twelve restrictions in the table, and it does not appear for the outcome that the spending weights are built to explain.

Three features of the problem limit what such a test can establish. The first is the lesson of Section 4: the readings differ only when asset prices move in unusual proportions, so the variation in

Table 3: Does the Composition of a Tightening Help Predict Activity?

	n	Market reading, β	Market reading sufficient: $\gamma = 0$	FRB/US reading sufficient: $\gamma - \beta = 0$
Real GDP growth over the next four quarters				
1976–2026	186	−0.16 [−0.47, 0.16]	0.87 [−0.78, 2.51]	1.03 [−0.39, 2.44]
1996–2026	106	0.31 [−0.13, 0.75]	1.81 [−0.76, 4.38]	1.50 [−0.70, 3.71]
Excluding 2008–15	146	−0.33 [−0.61, −0.04]	0.26 [−1.21, 1.72]	0.58 [−0.69, 1.86]
Change in the unemployment gap over the next 12 months				
1976–2026	556	0.19 [0.05, 0.33]	−0.26 [−0.94, 0.43]	−0.45 [−1.01, 0.11]
1996–2026	318	0.01 [−0.21, 0.22]	−1.19 [−2.50, 0.13]	−1.19 [−2.31, −0.07]*
Excluding 2008–15	436	0.29 [0.15, 0.43]	0.32 [−0.13, 0.77]	0.03 [−0.30, 0.37]

Notes: Local projections of four-quarter-ahead real GDP growth (quarterly) and of the 12-month-ahead change in the unemployment gap (monthly) on own lags and $\beta M + \gamma D$, where M is the 12-month change of the futures-factor market reading and D the FRB/US reading of the same asset changes less M, both on the common scale. Brackets are 90 percent intervals (HAC, joint covariance of the coefficients); * marks a restriction rejected at the 10 percent level. If the market reading is a sufficient summary, $\gamma = 0$; if the FRB/US reading $M + D$ is, $\gamma = \beta$. The market weights are treated as known. A failure to reject is not evidence for either reading.

D_t that is independent of M_t is small and concentrated in a few episodes. In the quarterly sample the standard deviation of the market reading is 1.66 points and that of composition is 0.43. The two are negatively correlated (−0.66). The part of composition that is unrelated to the market reading has a standard deviation of 0.32, a fifth of the variation in the reading itself. The regression is asked to detect the effect of a small regressor in the presence of a large and correlated one. The second is that the sample contains five recessions, and the largest movements in composition coincide with two of them. The third is that asset prices are forward-looking. A movement in equity prices that reflects news about future earnings predicts activity for reasons unrelated to the effect of equity prices on spending, and a regression cannot tell the two apart.

The outcome evidence therefore bounds the claim of the paper and does not extend it. Realized activity does not show that one way of translating financial movements into points of policy is right. A policymaker has to choose a calibration knowing that the data will not adjudicate.

6 Scope and Conclusion

6.1 What Is Held Fixed, What Is Estimated and What the Intervals Cover

Three features of the design define the scope of the results. First, the four portfolios are a defined set of U.S. calibrations on three asset prices. The FRB/US weights come from one model, one horizon and one experiment. The published GS weights are applied to series that they were not built on, with their policy-rate and spread terms held fixed. Credit spreads, house prices and other variables that some indexes include are outside the comparison. Second, only the market weights are estimated here. The intervals come from resampling the 253 announcements and cover the sampling error of the market weights and of the response vector that defines the common scale. They treat the spending weights and the observed history as given, and they do not cover uncertainty about r^*. Third, the weights are estimated on announcements since 1996 and held fixed over the history.

Re-estimated on an expanding window, the weights on the 10-year yield and the dollar stay within the sampling error of the full-sample estimates; the weight on equities lies outside it in estimates between 2011 and early 2015 (Appendix E.2). If policy came to target financial conditions, the responses on which the market weights rest could change, as Hatzius et al. (2010) note for all such indexes.

6.2 Conclusion

What can a policymaker infer from a report that financial conditions have tightened by the equivalent of a quarter point? This paper compares four ways of making that translation for the same asset movements. Two ask how large a policy surprise the movement resembles, and two ask how large a change in the policy rate would have a comparable effect on GDP.

The two questions are different, and their answers have no reason to coincide. An asset's value as an indicator of policy news depends on how it behaves in announcement windows, and its importance for spending depends on consumption, investment and trade. The weights agree for movements in the proportions that policy news produces. For other movements they can differ, and they can agree by offsetting.

Knowing which question an index answers tells a reader little about which other index it will resemble. In these data the measure of policy news moves the market weights as much as the choice between market responses and spending effects moves the reading, the two spending calibrations differ from each other by as much, and the two closest readings come from different methods.

The direction of a change in financial conditions is a robust signal. Across the four calibrations and thirty years, readings that oppose each other by a meaningful amount occur in a handful of months, when the dollar and equity valuations move strongly in opposing directions. The size of a change is not robust. In 2008–09 the estimated tightening differed by more than half a point across calibrations that agreed completely on its direction, and in 2022–24 calibrations with different weights agreed on the size only because the differences offset. Subsequent activity does not say which calibration to prefer.

A reported quarter point of tightening should therefore come with two pieces of information: the calibration that produced it, and which asset prices moved. When yields, equities and the dollar move as they do around policy news, any calibration will do. When one of them moves alone, or two move against each other, the number depends on a choice that the data cannot make for the policymaker.

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Appendix A The Weights: a Derivation

Let the movements in the three asset prices be $x = \psi s + u$, where s is policy news, ψ the responses and u everything else, with covariance Σ. Let g_h be the effects of a persistent unit shift in each asset price on spending at horizon h. The FRB/US weights are proportional to g_h. The market weights solve the problem of recovering s from x: the generalized least squares estimator is $\hat{s} = w_M' x$ with $w_M = \Sigma^{-1}\psi/(\psi'\Sigma^{-1}\psi)$, which satisfies $w_M'\psi = 1$ and, by the Gauss–Markov theorem, minimizes the variance of the estimate among linear unbiased estimators of s when u has mean zero and covariance Σ and is uncorrelated with s. For any x the difference between the normalized readings is $D_h(x) = (w_{B,h} - w_M)'x = (w_{B,h} - w_M)'u$, because the policy part cancels. The readings coincide for every movement if and only if $w_{B,h} = w_M$, that is, if $g_h \propto \Sigma^{-1}\psi$: the criterion that recovers policy news precisely and the criterion that measures spending effects select the same weights only in that case. The variance of the difference is $(w_{B,h} - w_M)'\Sigma(w_{B,h} - w_M)$.

Observed movements need not be of the form $\psi s + u$ with a single policy factor, and the model's effects need not be those of a movement of the same shape. If all observed movements lay along the single direction ψ, every weight vector normalized to price ψ at one would give the same reading on them. The data would then identify the reading along that direction and leave the weights themselves undetermined. In the announcement windows u is not zero, and it is the variation in u that identifies Σ and hence the weights. What distinguishes the two weightings in practice is the exposure of the difference in weights to the movements that occur, which Table A.1 measures as a ratio to the market move.

Appendix B Construction of the Series

The three asset prices enter in three forms, each answering a problem of the one before. The headline results use the third.

Nominal. Rates as published. A nominal index treats a 10-year yield of 8 percent alike in 1981 and in 2000. That is reasonable only if inflation expectations and the neutral real rate are stable,

Table A.1: How Large is the Difference Between the Two Readings?

Movements	Mean $\lvert D\rvert$	Ratio to market move [90%]	Share above 0.25 (%)	Mean run (months)
Announcement windows, held out	0.010	0.19 [0.09, 0.37]	0	–
Monthly changes, 1996–2026	0.043	0.24 [0.14, 0.54]	0	1.0
Monthly changes, 1973–2026	0.063	0.27 [0.15, 0.51]	2	1.1
12-month changes, 1996–2026	0.26	0.33 [0.17, 0.63]	37	7.5
12-month changes, 1973–2026	0.33	0.32 [0.16, 0.54]	45	7.3

Notes: D is the FRB/US reading (four quarters, normalized so that a conventional policy surprise reads one) less the market reading, applied to the changes of the equity valuation, the 10-year yield and the dollar at the r^* step. Announcement windows: held-out events (weights re-estimated without each event). Ratio: mean $\lvert D\rvert$ over mean absolute market move; interval from bootstrap samples of the announcements and, for the monthly history, of 12- or 24-month blocks of months. A run is consecutive months with $\lvert D\rvert$ above 0.25.

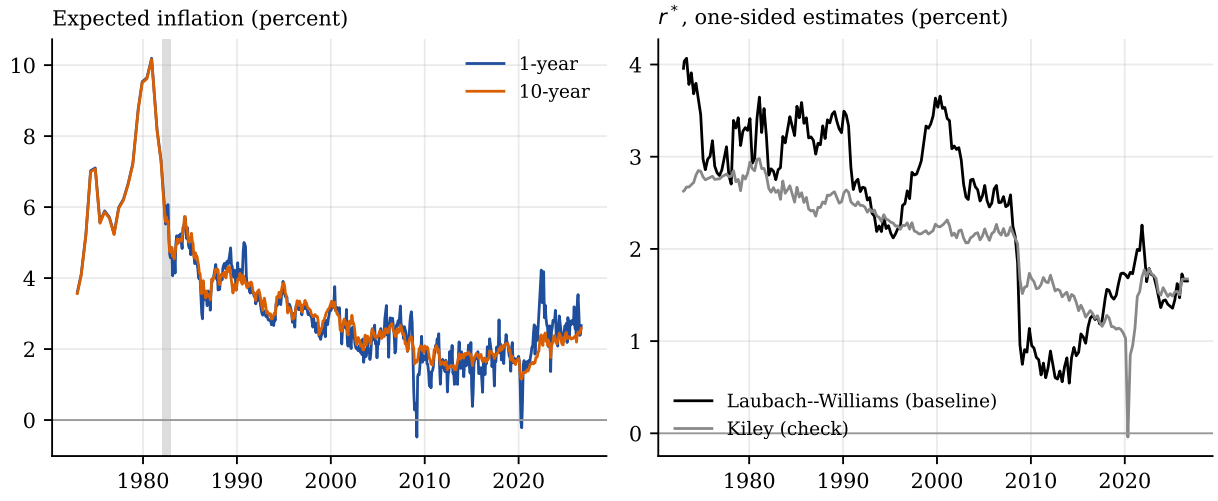


Figure B.1: Expected Inflation and r^*

Notes: Left: expected inflation, Livingston Survey before 1982 and Cleveland Fed model estimates after, shifted to the Cleveland level over 1982–84 and blended over 1982; shaded: the blending year. Right: one-sided, full-sample estimates of r^* from the Laubach–Williams model (baseline) and Kiley's model (check). Percent.

and neither was. Expected inflation over the next year averaged 8.43 percent in 1981 and is 2.66 percent today, and the Laubach–Williams estimate of r^* averaged 2.76 percent over 1990–2007 and 0.86 percent over 2009–2016 (Figure B.1).

Real. Each rate less expected inflation at a matched horizon: ten-year expected inflation for the 10-year yield. The dollar is the real broad index.

Relative to r^*. The real 10-year yield less r^*. Equity valuation enters through the earnings yield $E/P = 100/\text{CAPE}$, which is also an implicit real rate, as the gap $E/P - r^*$, converted to CAPE log points at the sample-average earnings yield so that its weight on CAPE is unchanged. The baseline uses the one-sided, full-sample Laubach–Williams estimate. Each variable is measured relative to its 1973–2026 average, so r^* matters through its movements over time.

The transformation matters for levels and not for the difference between the two readings. The ratio of the mean absolute difference to the mean absolute market move in 12-month changes, 1996–2026, is 0.33 (interval 0.18 to 0.68) with nominal assets and 0.33 relative to r^*.

Data. All series are monthly averages from January 1973 to September 2026, from the public sources in Table B.1. The BBB spread is Moody's Baa yield less the 10-year yield. Months in which the CPI and unemployment releases are missing (October 2025) are interpolated. The unemployment gap is the unemployment rate less the Congressional Budget Office's natural rate, placed in the middle month of each quarter and interpolated. Expected inflation is the Cleveland Fed's estimate at one and ten years from 1982 and the Livingston Survey's 12-month CPI expectation before, shifted to the Cleveland level by the mean difference over 1982–84 and blended over 1982. CAPE is Shiller's series spliced over 1990–92 in logs to the monthly average of a daily series. The real broad

dollar is spliced in 2006 in logs, as are the nominal dollar and the S&P 500.

Market index. Policy news is the first principal component of MP1, MP2, ED2, ED3 and ED4 in the monetary-event windows, scaled by its regression on the two-year yield change so that one unit moves the two-year yield by one percentage point. Unscheduled events with no change in MP1 and March–April 2020 are dropped. Each of the 10-year yield change, the S&P 500 return and the dollar (DXY) return in the window is regressed on the factor. The residual covariance Σ of the three regressions gives $w = \Sigma^{-1}\psi/(\psi'\Sigma^{-1}\psi)$. The S&P 500 and DXY weights apply to 100 times the log of CAPE and of the real broad dollar, as percent changes in the window and in the index levels agree to first order. The weights bootstrap resamples announcements with replacement and recomputes the policy-news factor and the weights each time; draws use the seed in `lib/config.py`.

FRB/US index. The weights are the cumulative GDP responses at four quarters to a persistent unit shift in each variable, simulated in FRB/US with VAR-based expectations, divided by the response to a persistent one percentage point rise in the funds rate. The weight on the policy rate is its direct effect with expectations of future rates held fixed. House prices enter as 100 times the log of the price-to-rent ratio relative to its trailing 15-year average. The expenditure decomposition of Figure 3 uses the model's own identities for consumption, residential and business fixed investment, inventories, government, exports and imports, weighted by their shares in GDP, with a residual below 10^{-7} percentage points; it is run with the Board's FRB/US package and the data vintage of October 2026, downloaded by step 00 into a gitignored directory, with a Python 3.11 environment built on first use. The policy-rate experiment shocks the inertial-Taylor-rule equation in the first quarter and is scaled (to 2.14 units of the add factor) so that the average simulated funds rate over eight quarters rises by one point; GDP at four quarters is linear in the size of the shock to within 1.6 percent. For the decay experiments of Section D.4, each variable's shock decays geometrically with the half-life stated, and the response is computed in the model by linearity from the temporary-shock responses.

Policy-news definitions. The two-year definition replaces the futures factor by the change in the two-year Treasury yield in the same window, with the same regressions and the same bootstrap of announcements; the draws for different definitions are paired by keeping, for each resample, the announcements that fall in each subsample, so the subsample size varies by draw. The announcement bootstrap uses 5,000 draws for every weights-based interval; the block bootstraps of the VAR and the outcome regressions use 500. The FRB/US weights are held fixed.

Forecasting data. The regressions use local projections of the 12-month-ahead change in the unemployment gap (four-quarter-ahead GDP growth) with HAC standard errors of 18 lags.

Appendix C The Four Calibrations: Additional Results

C.1 The Market Weights and Their Precision

Table C.1 gives the market weights under the baseline. A policy move that raises the two-year yield by one point raises the 10-year yield by 0.53 points, lowers equity valuation by 7.78 percent and

Table B.1: Data Sources

Series	Frequency	Source
Federal funds rate; 10-year yield; Baa yield; 30-year mortgage rate	Monthly	FRED FEDFUNDS, GS10, BAA, MORTGAGE30US
Real broad dollar; nominal broad dollar	Monthly	FRED RTWEXBGS, TWEXBPA; DTWEXBGS, TWEXBMTH (spliced in 2006)
CPI; CPI rent; Case–Shiller home prices; S&P 500	Monthly	FRED CPIAUCSL, CUUR0000SEHA, CSUSHPISA, SP500; Shiller's data before the FRED start
Unemployment rate; natural rate of unemployment; real GDP	Monthly; quarterly	FRED UNRATE; CBO, FRED NROU; GDPC1
Expected inflation, 1 and 10 years	Monthly	Cleveland Fed, FRED EXPINF1YR, EXPINF10YR; before 1982, Livingston Survey (Philadelphia Fed)
r^*, Laubach–Williams	Quarterly	New York Fed, one-sided estimates
r^*, Kiley	Quarterly	One-sided estimates of the author
CAPE	Monthly	Shiller, spliced to the monthly average of a daily CAPE constructed by the repository's daily workflow
FRB/US simulations	Quarterly	Run for this paper's companion work with the public FRB/US package
Futures, yields, S&P 500 and DXY in the window	Event	U.S. Monetary Policy Event-Study Database, "monetary events" (Acosta et al., 2025)

raises the real dollar by 4.22 percent. Each variable carries a substantial part of the policy move, with shares of 0.33 for the 10-year yield, 0.40 for equity valuation and 0.28 for the dollar.

The weights come from regressions on 253 announcements. The weight on the 10-year yield has a 90 percent interval from 0.15 to 1.11 around 0.61. The interval for equity valuation runs from -0.07 to -0.03, and for the dollar from 0.02 to 0.11. The FRB/US weights on all three lie inside these intervals, the equity weight near the edge: it is lower in absolute value than 5 percent of the bootstrap draws, about 14 percent in a two-sided test. The data do not reject that the model weights equal the market weights. This is a statement about power, not a finding that they are equal, and it treats the FRB/US weights as fixed. The intervals in the text come from the

Table C.1: Market Index: Responses to Policy News and Weights

	Response	Weight	90% interval	Share of policy move
10-year Treasury yield (pp)	0.53 (0.06)	0.609	[0.147, 1.114]	0.33
Equity valuation, 100 ln CAPE (percent)	−7.78 (0.93)	−0.051	[−0.068, −0.030]	0.40
Real broad dollar, 100 ln (percent)	4.22 (0.53)	0.065	[0.021, 0.112]	0.28

Notes: Announcement windows, 253 announcements, January 1996 to September 2026. Response: change in the window per unit of policy news, which moves the two-year yield by 1 percentage point (heteroskedasticity-robust standard errors in parentheses). Weight: one-factor generalized least squares, normalized so that the weights applied to the responses sum to 1. Interval: 5th and 95th percentiles of weights re-estimated on 5000 bootstrap samples of the announcements. Share: weight times response.

bootstrap of the announcements; the contrast in composition in Section D.4 is the ratio of ratios, whose interval includes one at two quarters.

Readings of the two movements. Table D.7 in the appendix applies the weights, normalized for each signal, to the movements of Figure 1. The market-implied reading rates the equity decline and the dollar appreciation 0.00 apart with the futures factor and 0.14 apart with the two-year yield. The model rates the dollar movement 0.24 points tighter than the equity movement. The model less market contrast is 0.243 (90 percent interval 0.003 to 0.448) for the baseline and 0.107 (-0.079 to 0.287) for the two-year yield. On common draws, switching from the futures factor to the two-year yield lowers the contrast by 14 basis points of the two-year yield (-0.14, interval -0.27 to 0.03). The model gives the dollar more weight relative to equities than the market does in the point estimates under both measures, and the interval for the difference includes zero under the two-year yield.

Proportionality. Whether the model's dollar-to-equity weight ratio equals the market's is a question about relative weights and not about size. The cross-product $c = w_{B,\$} w_{M,E} - w_{B,E} w_{M,\$}$ is zero if and only if they are equal. Its 90 percent interval excludes zero for the futures factor (-0.0032, -0.0058 to -0.0004) and includes it for the two-year yield (-0.0005, -0.0029 to 0.0015). The ratio of the model's ratio to the market's, which is 2.53 for the baseline and 1.22 for the two-year yield (interval 0.48 to 3.05), tests the same hypothesis on a scale-free ratio, but its denominator, the market's equity effect, is small and imprecise. The two sample checks, announcements through 2019 and outside the lower-bound years, give contrasts of 0.349 (0.036 to 0.565) and 0.232 (0.021 to 0.405); Appendix E.1 reports them. The intervals are not adjusted for the number of definitions.

C.2 The Spending Weights

Table D.9 compares the market weights with the FRB/US weights. The two sets agree on the 10-year yield, where the FRB/US weight is 0.92 times the market weight. The model puts less weight on equity valuation (0.60 times) and more on the dollar (1.51 times). On a policy move the difference does not matter, because the policy part cancels in equation (5). On a departure from the usual response, the model gives a fall in equity prices its estimated effect on output, and the market index treats it as policy-equivalent.

The equity movement reduces GDP by -0.070 percent after four quarters, and domestic demand accounts for it. Consumption supplies 92 percent of the effect, through wealth and the cost of capital, and business fixed investment 27 percent. Domestic demand in all supplies 120 percent, and net exports offset 33 percent of it, because weaker demand reduces imports; inventories supply 13 percent and government -1 percent, so that the shares sum to 100. The dollar movement reduces GDP by -0.177 percent, and net exports account for 94 percent of it: exports 73 percent and imports 21 percent. Domestic demand contributes 0.6 percent, inventories 13 percent and government -7 percent of the GDP effect. With a half-life of four quarters the shares are similar (consumption 91 percent of the equity effect and net exports 91 percent of the dollar effect). The two weights therefore reflect two different channels, and the dollar's larger weight is a statement

about trade. It carries no implication about how persistent the dollar is, which is the assumption of the experiment.

The experiments need two mappings from the market series to the model. The equity experiment shocks the required return on equity, and household stock wealth falls with it in the same simulation, so valuation and wealth move jointly; the 5 log point fall in valuation is entered as a 5 percent fall in stock wealth on impact. The market side measures equity by the S&P 500 return in the announcement window, which gives the weight on equity valuation, and applies that weight to monthly changes in the earnings-yield gap converted to CAPE log points (Appendix B); a one percent window return is treated as one log point of valuation, with earnings unchanged. The dollar experiment raises the model's real exchange rate by one percent per unit, and the market side measures the dollar by the DXY in the window and applies the weight to the log of the broad real dollar. DXY is a nominal index of six currencies and the model's dollar is a trade-weighted real index. I treat them as the same unit, and over a window or a month the nominal moves dominate.

A check on the spending side. The multipliers of the Board's FCI-G offer a check, with the caveat that they share a lineage with FRB/US, so agreement measures robustness within a family. Per 1 percent change in each variable, the FCI-G dollar multiplier is 2.25 times its equity multiplier, against 3.22 in FRB/US (these ratios use the multipliers per 1 percent, not the sizes of the movements of Figure 1, for which the FRB/US ratio is 2.51). The ratio of ratios against the market is 1.76 under the baseline signal and 0.85 under the two-year yield. FCI-G corroborates the direction under the futures factors and not under the two-year yield. A genuinely independent check would use another model with documented asset-specific multipliers.

C.3 Distances, Episodes and Months

Table C.2 gives the six pairwise distances behind Table 2, Table C.3 the average readings in three episodes, Table C.4 the judgments across the four readings with their intervals, and Table C.5 the months of materially opposing readings. These are rankings of point estimates. Over 5,000 bootstrap draws of the announcements, the futures-factor to GS distance has a 90 percent interval of 0.07 to 0.57 of the typical move. A distance cannot be negative, so when two sets of weights nearly coincide, sampling error in one of them tends to raise the distance; the point estimate sits near the bottom of its interval for that reason. The interval for the FRB/US to GS distance, 0.30 to 0.44, is narrow because neither set of weights is estimated here; only the common normalization varies. The paired difference between the futures-factor to FRB/US distance and the distance between the two market calibrations is −3 basis points (−19 to 27): the two are comparable in this sample, and the data cannot rank them.

The exceptions. Table C.5 lists the 8 months of materially opposing readings. They fall in four periods, 1997–98, 2003, 2008 and 2015, and show two patterns. In 6 months, in late 1997, the first half of 1998 and early 2015, FRB/US alone reads a tightening of at least 25 basis points while the others read an easing. In each of those months the dollar had risen by more than 10 percent over the year, valuations had also risen by more than 10 percent and the 10-year yield had fallen.

Table C.2: How Far Apart Are the Four Readings?

	Market, futures factor	Published GS weights	Market, two-year yield	FRB/US
Market, futures factor	–	7 (0.09)	28 (0.37)	26 (0.33)
Published GS weights	7 (0.09)	–	34 (0.45)	28 (0.36)
Market, two-year yield	28 (0.37)	34 (0.45)	–	23 (0.30)
FRB/US	26 (0.33)	28 (0.36)	23 (0.30)	–

Notes: Mean absolute difference between two readings of the same 12-month changes in the three assets relative to r^*, monthly, January 1996 to September 2026 (369 months), in basis points of the two-year yield; in parentheses, the same divided by the mean absolute reading of the market index with the futures factor (the common denominator). Every portfolio is normalized to read one on the futures-factor response to policy news, so that the distances reflect relative weights. Published GS weights: the U.S. weights on the 10-year yield, equities and the dollar, applied to the same series, with the GS policy-rate and spread terms held fixed. Intervals over 5,000 bootstrap draws of the announcements are in the text.

Table C.3: Four Readings of Three Episodes

Episode (months)	Market, futures factor	Published GS weights	Market, two-year yield	FRB/US	Mean range	Opposing months
2008–09 (January 2008 to December 2009)	2.05	2.17	1.33	1.40	1.06	1
1998–2001 (January 1998 to December 2001)	−0.06	−0.03	−0.07	0.16	0.49	3
2022–24 (March 2022 to December 2024)	1.02	0.97	1.21	1.01	0.32	0

Notes: Average over the months of each episode of the 12-month change in financial conditions under each portfolio, in points on the common two-year-yield scale (normalized to the futures-factor response). Mean range: average over the months of the largest less the smallest reading. Opposing months: months with at least one reading at least 0.25 above zero and one at least 0.25 below. The readings use the 10-year yield, equity valuation and the dollar only; corporate spreads, which the published GS index and the full FRB/US index contain, are held fixed.

Table C.4: How Often the Four Readings Give Different Judgments

	Percent of months	90% interval	Months
All four readings have the same sign	81.6	[70.2, 82.9]	301
At least two materially opposing readings, $t = 10$ bp	10.3	[8.4, 19.8]	38
At least two materially opposing readings, $t = 25$ bp	2.2	[1.6, 8.1]	8
Four-reading range at least 10 bp	92.4	[87.5, 97.6]	341
Four-reading range at least 25 bp	70.5	[58.5, 88.9]	260
Market (futures factor) and FRB/US materially opposing, $t = 25$ bp	2.2	[0.0, 5.4]	8

Notes: Monthly 12-month changes in the three assets relative to r^*, January 1996 to September 2026 (369 months, overlapping), read with the four portfolios on the common two-year-yield scale. Materially opposing: at least one reading at least t above zero and at least one at least t below zero. Four-reading range: the largest less the smallest of the four readings. Intervals: 5th and 95th percentiles over 5,000 bootstrap draws of the announcements (market weights and the common response vector redrawn; FRB/US and GS weights fixed), conditional on the observed history.

Table C.5: The Months of Materially Opposing Readings

Month	Market, futures factor	Published GS weights	Market, two-year yield	FRB/US
Dec 1997	-0.43^{-}	-0.40^{-}	-0.24	0.27^{+}
Jan 1998	-0.49^{-}	-0.41^{-}	-0.42^{-}	0.28^{+}
Jun 1998	-0.33^{-}	-0.27^{-}	-0.23	0.37^{+}
Jul 1998	-0.39^{-}	-0.35^{-}	-0.26^{-}	0.29^{+}
May 2003	0.41^{+}	0.53^{+}	-0.28^{-}	-0.30^{-}
Jun 2008	0.46^{+}	0.56^{+}	-0.18	-0.26^{-}
Mar 2015	-0.37^{-}	-0.30^{-}	-0.37^{-}	0.26^{+}
Apr 2015	-0.37^{-}	-0.30^{-}	-0.35^{-}	0.25^{+}

Notes: Twelve-month change in financial conditions under each portfolio, in points on the common two-year-yield scale (normalized to the futures-factor response), in the months in which at least one reading is at least 0.25 above zero ($^{+}$) and at least one at least 0.25 below zero ($^{-}$).

FRB/US gives the dollar the largest weight relative to equities, so the appreciation dominates its reading; the others give more weight to the rise in valuations and the fall in yields. In the other 2 months, May 2003 and June 2008, the futures-factor and GS readings are tight while FRB/US is easy. Valuations had fallen by more than 20 percent over the year and the dollar had depreciated, and those two portfolios weight equities most heavily relative to the dollar.

On common draws, the futures-factor to GS distance less the nearest other distance has an interval that includes zero under each of the three sets of series of Table 2 (−16 to 28 basis points on the paper's series), so the identity of the closest pair is a point-estimate ranking.

C.4 The Futures-Factor and FRB/US Pair in Two Episodes

Figure C.1 follows the pair month by month in 1998–2001 and 2022–24, with the contribution of each asset to the difference D between the FRB/US reading and the market reading. **1998–2001.** For the pair of the futures-factor market index and FRB/US, averaged over the episode, the market-implied reading of the 12-month changes was slightly easing, at −0.06 points (interval −0.37 to 0.27), and the spending-equivalent reading was slightly tightening, at 0.16 (0.14 to 0.17). The difference of 0.22 (−0.12 to 0.53) is a point estimate; its interval includes zero, and so do the intervals for the separate contributions of the dollar (−0.05 to 0.36) and of equities (0.00 to 0.10). The signs differ in the averages and in 7 of the 48 months, and the monthly paths show what the average conceals (Figure C.1). The difference was largest in the 12-month changes ending in the first half of 1998, at 0.76 points in January 1998, because those changes include the dollar's appreciation and the rise in valuations of 1997, and the model gives the dollar more weight and equities less. It shrank through 1999 and 2000 and turned negative in the second half of 2001 as valuations fell. The disagreement was therefore mostly about size and not direction: the signs differ in only 7 of 48 months, but the readings differ by at least a quarter point in 21. The dollar appreciated by 4.18 log points a year on average, which adds 0.16 to D; equity valuations rose by 2.70 log points a year, which adds

0.05. Unemployment rose by 0.35 points over the following year, against forecasts of 0.67 from the market-implied reading and 0.78 from the spending-equivalent reading.

2022–24. For the futures-factor pair, the two readings of the 12-month change averaged 1.02 points (interval 0.69 to 1.35) and 1.01 (0.90 to 1.13). Both movements tightened conditions. Equity valuations fell by 5.04 log points a year, and the dollar appreciated by 2.85. Because the model gives equities less weight than the event study, the fall in equities makes the spending-equivalent reading less tight than the market-implied one, and contributes -0.10 to D. Because the model gives the dollar more weight, the appreciation makes the spending-equivalent reading tighter, and contributes 0.11. The two contributions to the disagreement cancel, so that D averages -0.01 (-0.27 to 0.26), although each asset tightened conditions. The monthly paths show that the cancellation is not exact at every date: D ranged from about -0.3 to $+0.5$ within the episode, and it exceeded 0.25 in absolute value in 6 of 34 months. The baseline cancellation is specific to that definition of policy news: with the two-year yield as the signal, D is -0.17 and the market-implied reading is higher, and without the lower-bound years D is -0.19 (Section 3.1). A policymaker would have described a sizable tightening under every one of them. The full indexes differ in average level by 0.96 points (interval 0.00 to 1.76), with the equity weights (0.54) and the dollar (0.52) both contributing, and that difference does not affect the readings of the year's tightening. Unemployment rose by 0.27 points over the following year, against forecasts of 0.65 from the market-implied reading and 0.66 from the spending-equivalent reading. Both weightings overpredicted the rise, so the weights do not explain the miss.

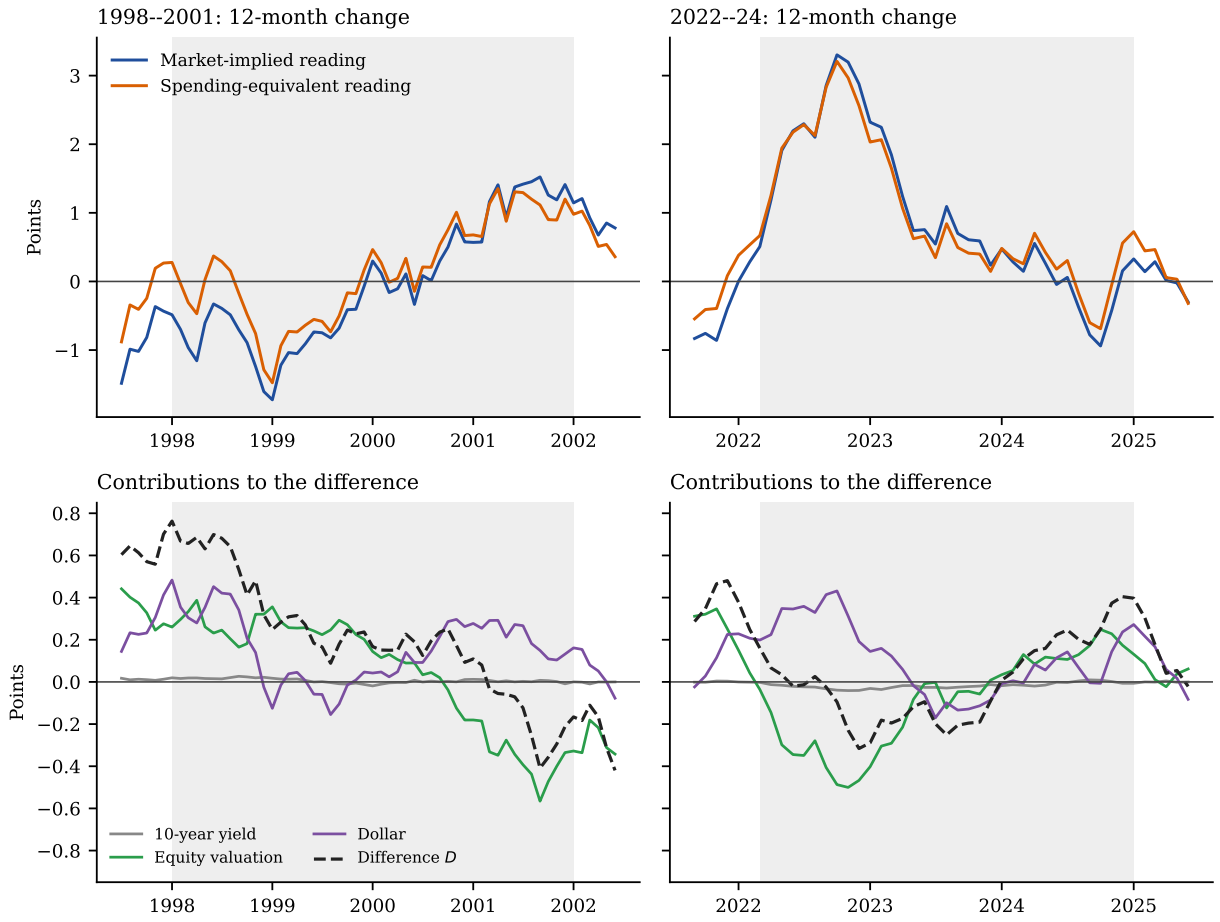


Figure C.1: Month by Month: Readings and the Sources of Their Difference

Notes: Top: 12-month change of the market-implied and spending-equivalent readings on the same three assets relative to r^*. Bottom: each asset's contribution to D, the weight difference times its 12-month change, and D itself (dashed). The shaded area is the episode.

Appendix D The Market and FRB/US Pair

D.1 Across Horizons

The pair that the comparison usually features, the futures-factor market index and FRB/US, can be followed at horizons from an announcement window to a year. On the 253 announcements, with the weights re-estimated without the event being evaluated, the ratio in intraday windows is 0.19 (interval 0.09 to 0.37), and the 90th percentile of the difference is 0.023 against a mean absolute market move of 0.053. These are the windows in which the weights are estimated, with each event held out.

The ratio is larger in daily data than in the intraday windows and does not shrink at longer horizons. On daily changes of the same three assets, 1996–2026, it is 0.30 on announcement days and 0.32 on other days; at one week it is 0.27 and 0.29, and at 22 trading days 0.27 and 0.25 (Table D.1). No difference between announcement and other windows is distinguishable from zero. Announcement days contain a great deal of financial news besides the announcement. The fitted response to policy news is a particular combination of asset prices, and the difference over a whole announcement day looks like the difference on other days.

For windows that end on any day, the ratio is 0.32, 0.29, 0.27, 0.30 and 0.34 at 1, 5, 22, 66 and 252 trading days. A window of 22 or more days contains announcements in any case; removing the moves on announcement days gives 0.33, 0.29, 0.28, 0.32 and 0.35, within 0.02 of those. In 12-month changes of the monthly series, the ratio is 0.33 (interval 0.17 to 0.63), and the difference exceeds 0.25 in 37 percent of months (Figure D.1). The months overlap, so runs in this series are not independent evidence of persistence. The statistic is a sample summary with wide intervals. Roughly 0.25 to 0.33 describes the baseline across horizons from a day to a year, and the ratio is 0.36 to 0.42 under the other definitions of policy news (Table E.3).

Sampling noise. How large is that relative to measurement noise in the market weights themselves? The same statistic computed on bootstrap draws of the baseline market weights against their point estimates has a median of 0.25 (5th to 95th percentile 0.08 to 0.51), and the baseline market weights against those of the two-year definition give 0.37 (0.15 through 2019 and 0.28 without

Table D.1: The Difference Between the Readings on Announcement Days and Other Days

Horizon	Announcement windows [90%]	n	Other windows [90%]	n
One day	0.30 [0.17, 0.56]	251	0.32 [0.18, 0.61]	7339
One week	0.27 [0.16, 0.57]	251	0.29 [0.16, 0.59]	1265
22 trading days	0.27 [0.15, 0.56]	248	0.25 [0.14, 0.52]	98

Notes: Mean absolute difference D between the normalized FRB/US reading and the market reading, as a ratio to the mean absolute market move, on the same three assets, 1996–2026 (daily 10-year yield, nominal broad dollar and CAPE; the nominal dollar and no adjustment for r^* stand in for the real dollar and the r^* transformation). Announcement windows end on a scheduled FOMC announcement date; other windows are non-overlapping and contain none. Intervals from joint bootstrap samples of the windows and of the announcements that set the weights.

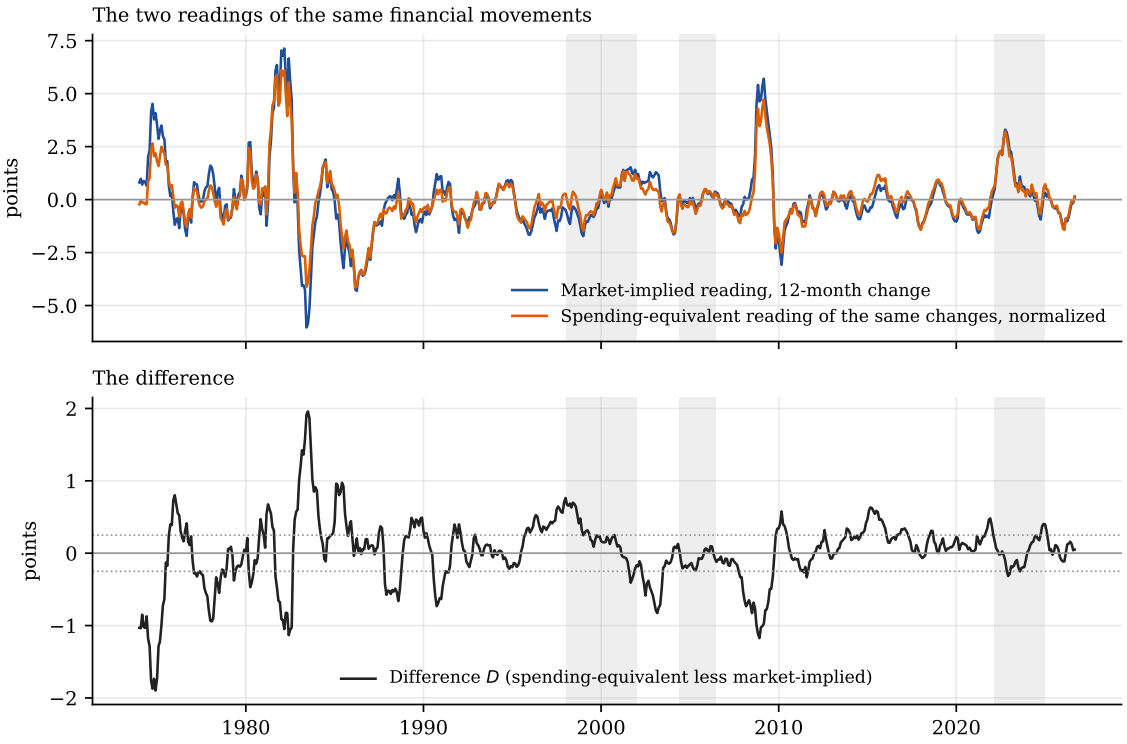


Figure D.1: Two Readings of the Same Financial Movements

Notes: Monthly, 12-month changes of the equity valuation, the 10-year yield and the dollar relative to r^*: the 10-year yield less ten-year expected inflation less r^*, the earnings-yield gap to r^* converted to CAPE log points, and the real dollar (Appendix B). Top: the change in the market index and the spending-equivalent reading of the same changes, normalized so that a conventional policy surprise reads one. Bottom: the difference D, FRB/US less market, with dotted lines at ±0.25. Shaded: 1998–2001, 2004–06 and 2022–24.

the lower-bound years), against 0.33 for the spending-equivalent weights against the baseline market weights. The disagreement between the model and the market is therefore about as large as the disagreement between two market calibrations that differ only in the measure of policy news (0.37 against 0.33), and it is of the size of the sampling noise in the market weights. The sampling noise limits how finely the two calibrations can be told apart. It is a statement about precision and is separate from the differences across calibrations.

The intraday and daily figures do not isolate a source for the gap between 0.19 and 0.3. The intraday windows are narrow and the weights exclude the event evaluated. The daily comparison uses nominal assets and weights fitted on the same announcements (Appendix E). The comparison across horizons is the cleaner result: the disagreement does not shrink as the window lengthens.

D.2 Months, Episodes and Levels

Table D.2 counts the months in which the two readings of the pair agree or differ, Table D.3 gives their readings of three episodes in 12-month changes and in levels, and Table D.4 accounts for the

Table D.2: How Often the Two Readings Differ

	1996–2026	1998–2001	2022–24
Months	369	48	34
Threshold t = 0.10 point			
Same sign	323 (87.5%) [78, 94]	41 (85.4%) [75, 94]	33 (97.1%) [76, 100]
Opposite signs, at least one reading within t of zero	27 (7.3%) [5, 10]	3 (6.2%) [4, 12]	1 (2.9%) [0, 12]
Opposite signs, both readings at least t	19 (5.1%) [0, 13]	4 (8.3%) [0, 17]	0 (0.0%) [0, 15]
Readings differ by at least t	269 (72.9%) [48, 88]	41 (85.4%) [54, 96]	23 (67.6%) [35, 91]
Threshold t = 0.25 point			
Same sign	323 (87.5%) [78, 94]	41 (85.4%) [75, 94]	33 (97.1%) [76, 100]
Opposite signs, at least one reading within t of zero	38 (10.3%) [6, 17]	4 (8.3%) [6, 19]	1 (2.9%) [0, 21]
Opposite signs, both readings at least t	8 (2.2%) [0, 5]	3 (6.2%) [0, 6]	0 (0.0%) [0, 3]
Readings differ by at least t	135 (36.6%) [10, 70]	21 (43.8%) [15, 81]	6 (17.6%) [0, 74]

Notes: Monthly 12-month changes in the three market variables (relative to r^*), read with the baseline market weights and with the FRB/US weights on the market's scale (points of the two-year yield). Each cell gives the number of months, the share of months, and the 5th to 95th percentile of the share over announcement-bootstrap draws of the market weights (the 12-month changes are held fixed; months overlap). 1996–2026 is 369 months, 1998–2001 is 48 and 2022–24 is 34.

difference in average levels between the full FRB/US index and the market index, which also differ in coverage (the policy rate, spreads and house prices) and centering.

D.3 Trade-Sensitive and Domestic Expenditure

Section 3.2 traced the model's weights to channels: consumption and investment for equities, net exports for the dollar. If the weights describe those channels, composition might predict the components separately even where it does not predict the aggregates. I wrote two predictions from the channel logic before estimating the components: the coefficient on D is negative for the contribution of net exports to growth over the next four quarters (P1), and it is larger for consumption and private investment than for net exports (P2). The aggregate nulls of the preceding subsections were already known, so the exercise is a check on a prediction and not a clean pre-registered test. Table D.5 reports both.

In 1976–2026 the coefficient on D for net exports is -0.74 (-1.06 to -0.41; bootstrap interval -1.04 to -0.37), an effect of -0.79 percentage points of growth between the 5th and 95th percentiles of D. For consumption and private investment it is 1.71 (-0.29 to 3.70), and the difference has a bootstrap interval of 0.13 to 4.88. Both predictions hold in the baseline main sample. They do not survive further scrutiny as evidence for the dollar channel. In this sample D is dominated by equities: it correlates 0.86 with the 12-month change in equity valuation and 0.31 with the change in the dollar. Splitting D into the parts due to each asset, the equity part has a coefficient of -0.97 (-1.67 to -0.27) and the dollar part -0.52 (-1.20 to 0.16). Rising equities strengthen domestic demand and imports, which lowers the net-export contribution mechanically. With the three asset changes entered freely, the dollar's own coefficient is negative, as the dollar channel implies (-0.030, -0.050 to -0.011), and so is the equity coefficient (-0.010). On 1996–2026 the

Table D.3: Two Readings of Three Episodes

	1998–2001	2004–06	2022–24
Full indexes, average level (points)			
FRB/US index less market index [90% interval]	1.79 [0.33, 2.97]	−0.11 [−0.76, 0.41]	0.96 [0.00, 1.76]
of which: equity weights	1.32	0.67	0.54
Same three assets, 12-month change (points)			
Market-implied reading	−0.06	0.04	1.02
Spending-equivalent reading	0.16	−0.05	1.01
Difference D	0.22	−0.09	−0.01
Change in the unemployment gap over the next 12 months (points)			
Realized	0.35	−0.41	0.27
Forecast from the market-implied reading	0.67	0.13	0.65
Forecast from the spending-equivalent reading	0.78	0.10	0.66

Notes: Top block: difference in the average levels of the full FRB/US index (with house prices) and the market index relative to r^*, in points, and the part that comes from the equity weights (Table D.4). Middle block: 12-month changes of the market index and of the normalized FRB/US reading, computed on the same three assets and averaged over the episode; D is their difference. Bottom block: change in the unemployment gap over the following 12 months, averaged over the episode's origins, and recursive out-of-sample forecasts of it from the market reading and from the FRB/US reading (own lags included). A gap in levels can coexist with agreement in changes.

Table D.4: From the Market Index to the FRB/US Index: Shares of the Difference

Stage	Level, share of variance (%)	12-month change (%)
Equity weights	72 [0, 89]	53 [-4, 87]
Dollar weights	12 [-1, 38]	15 [-1, 34]
10-year weight	1 [-4, 51]	1 [-7, 55]
Normalization of the weights	5 [-4, 19]	4 [-2, 16]
Policy rate	-0 [-1, 0]	0 [-1, 2]
BBB spread	1 [-1, 8]	3 [-3, 15]
Mortgage spread	2 [-3, 15]	15 [-0, 43]
House prices	7 [-2, 34]	8 [0, 24]

Notes: Shares of the variance of the FRB/US index (with house prices) less the market index, relative to r^*, 1973–2026, attributed to each stage of a bridge between them: the weights on the three common assets, the normalization of the weights, then the contributions that only the FRB/US index has (policy rate, spreads, house prices); both indexes use the same inputs and centering. Intervals (5th to 95th percentile) from bootstrap samples of the announcements and of blocks of months.

Table D.5: Composition and the Expenditure Components of Growth

	n	Coefficient [90% HAC]	Bootstrap 90%	Effect, 5th to 95th percentile
Net exports, 1976–2026: index change β	199	−0.11 [−0.19, −0.02]	[−0.19, 0.00]	−0.11
Net exports, 1976–2026: composition γ	199	−0.74 [−1.06, −0.41]	[−1.04, −0.37]	−0.79
Consumption and private investment, 1976–2026: index change β	199	−0.19 [−0.54, 0.15]	[−0.52, 0.22]	−0.21
Consumption and private investment, 1976–2026: composition γ	199	1.71 [−0.29, 3.70]	[−0.35, 3.92]	1.83
Consumption and investment less net exports, 1976–2026: γ	199	2.44	[0.13, 4.88]	2.63
Net exports, 1996–2026: index change β	119	0.03 [−0.13, 0.19]	[−0.13, 0.24]	0.03
Net exports, 1996–2026: composition γ	119	−0.54 [−1.17, 0.09]	[−1.28, 0.10]	−0.48
Consumption and private investment, 1996–2026: index change β	119	−0.31 [−1.10, 0.48]	[−1.13, 0.72]	−0.27
Consumption and private investment, 1996–2026: composition γ	119	2.95 [−0.65, 6.55]	[−0.98, 6.11]	2.59
Consumption and investment less net exports, 1996–2026: γ	119	3.49	[−0.99, 7.17]	3.07

Notes: Local projections of the mean contribution to real GDP growth (percentage points, annualized) of net exports and of consumption plus private investment over the next four quarters on own lags, the 12-month change of the market index and the 12-month change of D (Section 5.1); quarterly origins, HAC standard errors with 6 lags, 8-quarter block bootstrap with common draws for the two outcomes. Predictions stated from the channel logic before the components were estimated: the coefficient on D is negative for net exports (P1) and larger for domestic demand than for net exports (P2). Effect: coefficient times the 5th-to-95th percentile range of D at a given index change.

Table D.6: Composition and Outcomes Under Each Definition of Policy News

	Unemployment, γ	$\gamma - \beta$	GDP growth, γ	$\gamma - \beta$	Net exports, γ
Futures factor	−0.26 [−0.94, 0.43]	−0.45 [−1.01, 0.11]	0.87 [−0.78, 2.51]	1.03 [−0.39, 2.44]	−0.74 [−1.06, −0.41]
Two-year yield	−0.10 [−0.50, 0.31]	−0.33 [−0.75, 0.09]	0.53 [−0.51, 1.57]	0.77 [−0.25, 1.78]	0.02 [−0.37, 0.41]
Factor, through 2019	−0.12 [−0.64, 0.39]	−0.31 [−0.71, 0.09]	0.54 [−0.67, 1.76]	0.71 [−0.29, 1.70]	−0.54 [−0.78, −0.30]
Factor, no lower bound	−0.19 [−0.57, 0.20]	−0.38 [−0.72, −0.03]	0.68 [−0.35, 1.72]	0.85 [−0.07, 1.77]	−0.24 [−0.55, 0.07]

Notes: Coefficient on the composition term D (and its difference from the coefficient on the market index change) in the regressions of Sections 5.1 and D.3, 1976–2026, with D and the market index change computed with the market weights of each definition of policy news and the model weights normalized by that definition's $b_4'\psi$; HAC 90 percent intervals.

signs are the same and the evidence is weaker: the net-export coefficient is −0.54 (−1.17 to 0.09). The result also depends on the definition of policy news (Table D.6): the coefficient on D for net exports is −0.74 for the baseline, −0.54 through 2019, −0.24 without the lower-bound years and 0.02 with the two-year yield, where it is indistinguishable from zero. Composition predicts net exports under the futures-factor definitions, mostly through equities. It does not provide clean evidence for the dollar channel of the model.

Table D.6 also repeats the unemployment and GDP regressions under each definition. The coefficient on composition is not significantly different from zero in any of them, and the model's restriction $\gamma = \beta$ is rejected at the 10 percent level for unemployment only without the lower-bound years.

D.4 Composition or Persistence?

The comparison confounds two things. A persistent dollar appreciation in FRB/US and the dollar's response to policy news are different experiments, so a difference in weights could reflect the initial composition of the movement or the path that follows it. Figure D.2 separates them with the same set of calculations under common decaying paths for the three variables, with half-lives of 8, 4 and

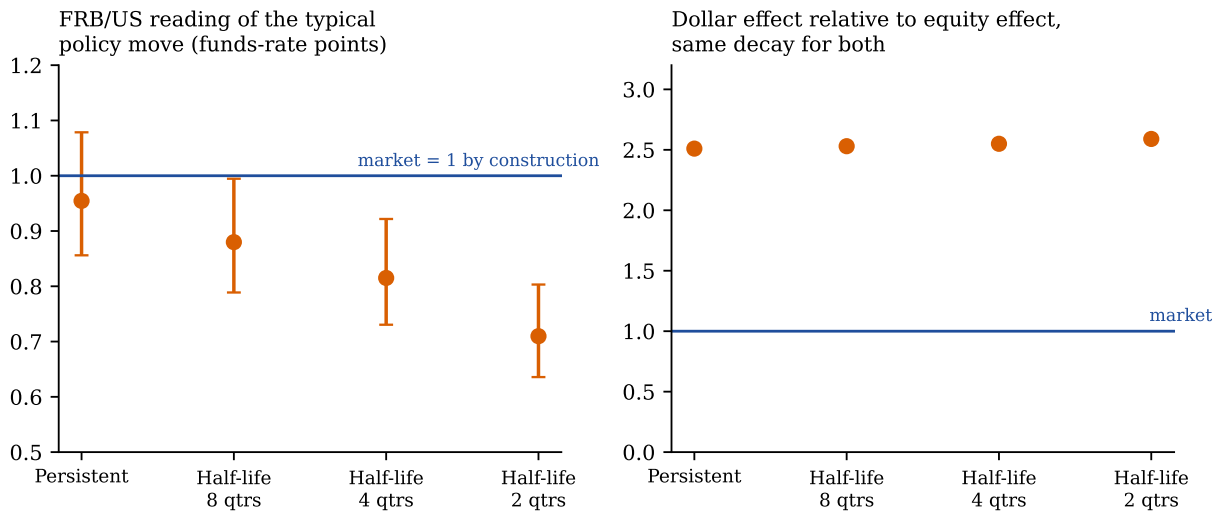


Figure D.2: Persistence Changes the Conversion, Not the Relative Weights

Notes: Left: the FRB/US weights at four quarters applied to the responses of the three variables to a unit of policy news, with the variables assumed to persist or to decay with the half-life shown; the bars are 90 percent intervals from bootstrap samples of the announcements. Right: the model's dollar effect over its equity effect, with the same decay for both; the line is the market index's ratio. The persistence of the variables' levels is a different object from the persistence of their responses to policy news; the half-lives of the levels since 1996 are 20 to 26 quarters, and the equity response to policy news does not decay within 20 trading days.

2 quarters.

Persistence changes how much tightening a policy move represents. The model's reading of the usual response falls from 0.954 with persistent paths to 0.82 (interval 0.73 to 0.92) with a half-life of four quarters and 0.71 with a half-life of two. Persistence does not change which movement counts more. With the same decay for the dollar and for equities, the model's dollar-to-equity ratio is 2.53, 2.55 and 2.59 at half-lives of 8, 4 and 2 quarters, against 1.00 for the market weights (interval 0.33 to 2.25). Different half-lives for the two assets give a wider range, 1.91 to 3.41, and none near one. Across horizons of two to eight quarters and these paths the model's reading of the usual response runs from 0.37 to 0.95. The persistent case is the closest to the data, since the half-lives of the levels of the three variables since 1996 are 20 to 26 quarters, so the range for decaying paths shows the sensitivity of the conversion and not a likely value. The contrast between the two weightings is a contrast in the effect on GDP per unit of each asset price. The definition of policy news (Section 3.1) moves the contrast by an amount at least as large.

D.5 Tables of the Weights

Table D.7: Two Measures of Policy News on the Same Announcements

	Futures factor	Two-year yield
Announcements	253	253
Market weight, 10-year yield	0.609	0.925
Market weight, equity valuation	−0.051	−0.027
Market weight, dollar	0.065	0.070
Market dollar-over-equity ratio [90%]	1.00 [0.33, 2.25]	2.06 [0.82, 5.19]
Readings of the two movements of Figure 1		
Equity valuation falls 5 log points, market (points)	0.256	0.133
Dollar appreciates 3.9 log points, market (points)	0.255	0.274
Equity valuation falls, FRB/US (points, normalized for each signal)	0.160	0.164
Dollar appreciates, FRB/US (points, normalized for each signal)	0.402	0.413
Dollar-minus-equity effect, market (points)	0.00	0.14
Dollar-minus-equity effect, FRB/US (points, normalized for each signal)	0.24	0.25
Model less market [90%]	0.243 [0.003, 0.448]	0.107 [−0.079, 0.287]
Ratio of ratios [90%]	2.53 [1.08, 7.41]	1.22 [0.48, 3.05]
Model reading of the usual response, $b_4'\psi$ [90%]	0.954 [0.856, 1.078]	0.930 [0.834, 1.048]

Notes: The market weights are estimated on the same 253 announcement windows with two measures of policy news: the first principal component of the futures-based factor of Nakamura and Steinsson (2018) (baseline) and the change in the two-year yield. The model weights are those of Table D.9 in both columns, normalized by $b_4'\psi$ for each measure. Effects are for the movements of Figure 1, in points of the two-year yield; the FRB/US effects are divided by $b_4'\psi$ computed with each signal's response vector, so this table's normalization changes with the signal, unlike the common futures-factor normalization of Tables 1 and C.2 (Table D.8 gives the FRB/US effects in funds-rate points). Intervals are 5th and 95th percentiles over 5,000 bootstrap samples of the announcements, on common draws. Table E.2 adds the two sample checks.

Table D.8: Two Financial Movements that the Market Index Rates Alike

	Market index	90% interval	FRB/US index
Equity valuation falls 5 log points	0.256	[0.152, 0.342]	0.153
Dollar appreciates 3.9 log points	0.255	[0.081, 0.438]	0.384
Dollar effect over equity effect	1.00	[0.33, 2.25]	2.51
Typical joint response to a unit of policy news	1.00		0.954 [0.86, 1.08]

Notes: Weights applied to the stated movements, with no change in the 10-year yield. Market index: two-year-yield points per movement, weights from announcement-window responses (Table C.1); the interval takes the 5th and 95th percentiles over bootstrap samples of the announcements. FRB/US index: funds-rate points; the effect on GDP after four quarters of a persistent shift in the variable, divided by the effect of a persistent one percentage point rise in the funds rate. Typical joint response: the index applied to the responses of the three variables to a unit of policy news (one by construction for the market index).

Table D.9: FRB/US and Market Weights on the Three Common Variables

	FRB/US weight	Market weight	90% interval	FRB/US response	Market response
10-year yield (pp)	0.563	0.609	[0.147, 1.114]	0.37	0.53
Equity valuation (percent)	−0.031	−0.051	[−0.068, −0.030]	−13.87	−7.78
Real dollar (percent)	0.098	0.065	[0.021, 0.112]	1.56	4.22

Notes: FRB/US weight: effect on GDP after four quarters of a persistent unit shift in the variable, divided by the effect of a persistent 1 percentage point rise in the federal funds rate (0.4616 percent of GDP); funds-rate points per unit. Market weight: two-year-yield points per unit (Table C.1); the interval is its bootstrap interval. FRB/US response: change in the variable four quarters after a persistent 1 percentage point funds-rate rise (VAR expectations). Market response: change in the announcement window per unit of policy news.

Table D.10: Agreement on Announcements and the Contrast in Composition

FRB/US horizon	Joint response to policy news [90%]	Normalized equity weight	Normalized dollar weight	Dollar over equity, FRB/US	Ratio of ratios [90%]
2 quarters	0.901 [0.81, 1.01]	−0.032	0.089	2.18	2.19 [0.94, 6.44]
4 quarters	0.954 [0.86, 1.08]	−0.032	0.103	2.51	2.53 [1.08, 7.41]
8 quarters	0.917 [0.82, 1.04]	−0.032	0.106	2.57	2.59 [1.11, 7.59]

Notes: FRB/US weights for GDP at the horizon shown, persistent-shift experiment, in funds-rate points; the joint response is the FRB/US weights applied to the responses to a unit of policy news (market index: one). Normalized weights: FRB/US weights divided by that response, so that both indexes read one on a conventional surprise; market weights are 0.609 (10-year yield), −0.051 (equity valuation) and 0.065 (dollar). Ratio of ratios: the FRB/US dollar-over-equity ratio divided by the market's (1.00); the interval comes from the bootstrap of the market weights, and the FRB/US weights are held fixed.

Appendix E Sensitivity

E.1 Other Policy-News Series and Event Samples

Table E.2 gives the weights, the readings of the two movements of Figure 1 and the contrasts under the two measures of policy news and the two sample checks, and Table E.1 re-estimates the weights with the two-year yield as the policy news, with announcements through 2019, and without the lower-bound years. The equity weight is −0.027, −0.061 and −0.039 against −0.051 in the baseline, and the conclusions about composition rest on the model's ratio of dollar to equity effects, not on any one set of market weights.

The episode readings of the futures-factor market index and FRB/US under each definition of policy news are in Table E.3.

Table E.1: Weights Under Other Policy-News Series and Event Samples

	Events	w 10-year	w equities	w dollar	ψ 10-year	ψ equities	ψ dollar
Baseline: Nakamura–Steinsson factor	253	0.609	−0.051	0.065	0.53	−7.78	4.22
Policy news is the 2-year yield change	253	0.925	−0.027	0.070	0.58	−6.09	4.22
Announcements through 2019	201	0.613	−0.061	0.053	0.56	−7.50	3.83
No announcements December 2008–November 2015	197	0.986	−0.039	0.058	0.49	−7.82	3.69

Notes: Weights w and responses ψ as in Table C.1, re-estimated on each sample or with the window change of the two-year yield as the policy-news measure. In each case the weights applied to the responses sum to 1.

Table E.2: Policy-News Definitions and Event Samples

	Futures factor	Two-year yield	Factor, through 2019	Factor, no lower bound
Announcements	253	253	201	197
Market weight, 10-year yield	0.609	0.925	0.613	0.986
Market weight, equity valuation	−0.051	−0.027	−0.061	−0.039
Market weight, dollar	0.065	0.070	0.053	0.058
Model reading of the usual response, $b_4'\psi$ [90%]	0.954 [0.856, 1.078]	0.930 [0.834, 1.048]	0.919 [0.805, 1.091]	0.876 [0.790, 0.983]
Market dollar-over-equity ratio [90%]	1.00 [0.33, 2.25]	2.06 [0.82, 5.19]	0.68 [0.09, 2.11]	1.16 [0.38, 2.89]
Ratio of ratios [90%]	2.53 [1.08, 7.41]	1.22 [0.48, 3.05]	3.70 [0.84, 16.30]	2.16 [0.85, 6.16]
Dollar-minus-equity effect, market (points)	0.00	0.14	−0.10	0.03
Dollar-minus-equity effect, FRB/US (points, normalized for each signal)	0.24	0.25	0.25	0.26
Model less market [90%]	0.243 [0.003, 0.448]	0.107 [−0.079, 0.287]	0.349 [0.036, 0.565]	0.232 [0.021, 0.405]
Mean $\lvert D\rvert$ over mean |market move|, 12-month changes	0.33	0.36	0.42	0.39

Notes: The market weights are re-estimated for each definition of policy news: the first principal component of the futures-based factor of Nakamura and Steinsson (2018) (baseline), the change in the two-year yield in the same 253 announcement windows, the factor through 2019, and the factor outside the lower-bound years. The model weights are those of Table D.9 in every column, normalized by $b_4'\psi$ for each definition. Effects are for the movements of Figure 1 (equity valuation down 5 log points, dollar up 3.9 log points), in points of the two-year yield; the FRB/US effects are divided by $b_4'\psi$ and so are on the common two-year-yield scale. Intervals are 5th and 95th percentiles over 5,000 bootstrap samples of the announcements, on common draws.

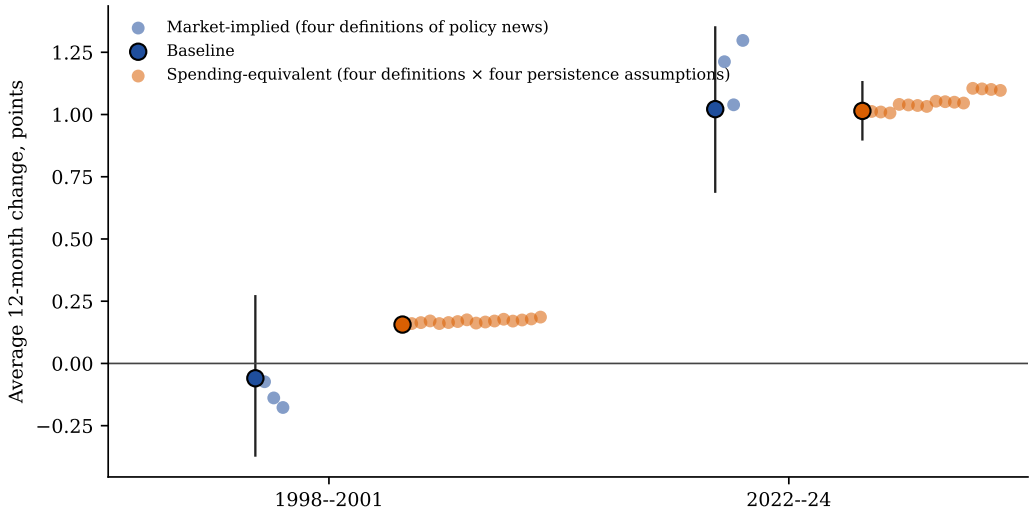


Figure E.1: Two Readings of Two Episodes under Every Measurement Choice

Notes: Average 12-month change over the months of the episode on the same three assets relative to r^*. Market-implied: the four definitions of policy news of Table E.2, in points of the two-year yield. Spending-equivalent: the same four definitions, which enter through the normalization, crossed with persistent paths and common half-lives of 8, 4 and 2 quarters; the FRB/US readings are divided by the model's reading of each definition's policy response and so are in points of the two-year yield, normalized for each definition. Large markers are the baseline. These are retrospective estimates.

Table E.3: Episode Readings Under Each Definition of Policy News

	Futures factor	Two-year yield	Factor, through 2019	Factor, no lower bound
1998–2001, average 12-month change				
Market-implied reading	−0.06	−0.07	−0.14	−0.18
Spending-equivalent reading	0.16	0.16	0.16	0.17
Difference D	0.22	0.23	0.30	0.35
2022–24, average 12-month change				
Market-implied reading	1.02	1.21	1.04	1.30
Spending-equivalent reading	1.01	1.04	1.05	1.11
Difference D	−0.01	−0.17	0.01	−0.19

Notes: Averages of the 12-month changes in the three assets relative to r^* over the months of each episode, read with the market weights of each definition (points of the two-year yield) and with the FRB/US weights normalized for the same definition (points on the common two-year-yield scale); D is the difference. Baseline intervals are in the text and in Table D.3.

The disagreement is similar under every definition: the ratio of the mean absolute difference to the mean absolute market move in 12-month changes is 0.33, 0.36, 0.42 and 0.39. The contrast in points excludes zero through 2019 and without the lower-bound years and includes it under the two-year yield, so that of the four definitions the interval excludes zero for three.

E.2 Weights That Move

The weights define the unit, so I hold them fixed and show how they would drift. Re-estimating on an expanding window of announcements from 2006 keeps the 10-year and dollar weights inside the 90 percent band of the full-sample estimate at every date after 2010; the equity weight is outside the full-sample 90 percent bootstrap interval in 32 monthly estimates between January 2011 and January 2015, by at most −0.073 against a band edge of −0.068 (Figure E.2). The overlapping monthly estimates are not 32 independent tests of instability. A 20-year rolling window fails after 2021, because it drops the announcements of 1996–2005. The 10-year weight is imprecise throughout. Apart from the equity weight in these years, the weights stay within their sampling error.

E.3 Further Limits of the Design

The weights come from announcements since 1996 and are applied from 1973; they assume that the responses reflect policy rather than information effects (Nakamura and Steinsson, 2018), and if policy came to target financial conditions the responses could change, the Lucas-critique concern that Hatzius et al. (2010) raise for all indexes. The FRB/US weights come from one model, at one horizon and in one experiment, and are held fixed in the intervals. The daily comparison of Appendix D.1 uses nominal assets, with the nominal broad dollar for the real dollar, no adjustment for r^* and weights fitted on the same announcements; the intraday comparison holds each event out. The sample has five recessions, two of them out of sample, and the 12-month outcomes overlap, so the counts of tests are descriptive and not adjusted for their number. The neutral rate is not neutral

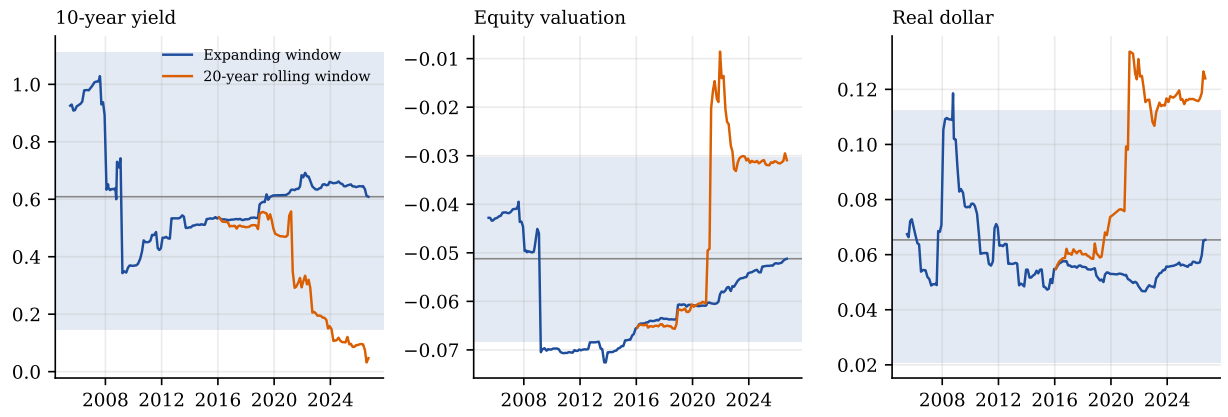


Figure E.2: The Weights on an Expanding and a Rolling Window

Notes: Weights of the market index re-estimated on announcements to each date (expanding window, from the 80th announcement) and on the latest 20 years (rolling). Shaded: 90 percent bootstrap interval of the full-sample weights; the horizontal line is the full-sample weight.

to outcomes, because the Laubach–Williams model uses output, inflation and unemployment.

E.4 The Policy-Rate Experiment in FRB/US

The model's own policy shock gives a funds-rate equivalent of 1.77 for the usual announcement mix against 0.95 for the announcement response. FRB/US can therefore assign a value near one to the observed announcement mix even though its own policy shock generates a different mix: heavier in equities and lighter in the dollar. The comparison is not a decomposition of how much of transmission the three assets omit, because the policy shock changes the path, the equity response, the channels that operate and the unit of normalization together. Appendix E.4 gives the experiment.

A policy shock in the model gives a comparison with the announcement response. I shock the equation of the inertial Taylor rule in the first quarter only, with the rule active afterwards and expectations from the model's VAR, and I scale the shock so that the average simulated funds rate over the next eight quarters rises by one point. FRB/US has no two-year yield, so that average is my proxy for the unit of policy news. Per unit of the proxy, the 10-year yield rises 1.33 times, equity wealth falls 4.01 times and the dollar rises 0.55 times as much as in the announcement response. GDP four quarters later falls by −0.82 percent, a funds-rate equivalent of 1.77, against 0.95 for the usual announcement response. Priced along their paths, the 10-year yield, equities and the dollar account for 20, 43 and 13 percent of the GDP effect of the model's policy shock, against 32, 25 and 44 percent for the announcement response, and 23 percent is left in the policy rate, spreads and house prices. The model's own policy transmission is heavier in equities and lighter in the dollar than the announcement response, and the difference in the funds-rate equivalent, 1.77 against 0.95, is the residual between the two experiments. The comparison has limits. The funds-rate average is a proxy, and a different proxy changes the unit. The equity fall on impact is large because stock

wealth in FRB/US follows expected funds rates through the required return. The exercise is the model's own policy shock, not a reproduction of an announcement. The reading of about one in Section 2.4 is therefore not the statement that announcements move asset prices as a policy shock does in the model; it is a statement about the combined effect on GDP of different mixes.